



Communications and Networks

Introduction and Information Theory

Sheng Zhou



Outline

- Course Information

- Introduction

- Information Theory



Course Information

■ Instructors

▶ Sheng Zhou (周盛)

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▶ Xiaofeng Zhong (钟晓峰)

- Office: Rohm 9-101, Phone: 62781389

- E-mail: zhongxf@tsinghua.edu.cn

■ Teaching Assistances

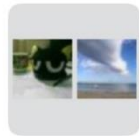
▶ 贾宇宽 (FIT楼3-328, jyk19980526@163.com)

▶ 李林涛 (954541436@qq.com)



Course Information (Cont.)

■ WeChat QR Code



通信与网络（英）2022 春课程群



该二维码 7 天内 (2 月 25 日前) 有效, 重新进入将更新



Course Information (Cont.)

■ Reference Books

- ▶ Simon Haykin, *Digital Communication Systems*, 1st ed., Wiley, 2013
- ▶ Ha Nguyen, Ed Shwedyk, *A First Course in Digital Communications*, 1st ed., 2009
- ▶ Rodger Ziemer, *Introduction to Digital Communication*, 2nd ed., Prentice-Hall, 2001
- ▶ Behrouz Forouzan, *Data Communications and Network*, 5th ed, McGraw-Hill, 2013
- ▶ And more....
- ▶ Please read the reference book **on the last slide of each chapter**

Experiments

■ Instructors

- ▶ Changyong Pan pcy@tsinghua.edu.cn 83998
- ▶ Jintao Wang wangjintao@tsinghua.edu.cn 96300

■ Experiments

- ▶ PCM encoder
- ▶ ASK modulator and demodulator
- ▶ BPSK transmission

■ Location

- ▶ Central Main Building 910、916

■ Schedules

- ▶ Will be announced later (depending on the situation)





Credits

■ Homework ——— 10%

▶ Overdue will incur credit discount

■ Experiments ——— 10%

▶ Overdue will incur credit discount

■ Midterm exercise ——— 20%

▶ Details about the date and form will be announced later depending on the situation

■ Final exam ——— 60%



Outline

- Course Information

- Introduction

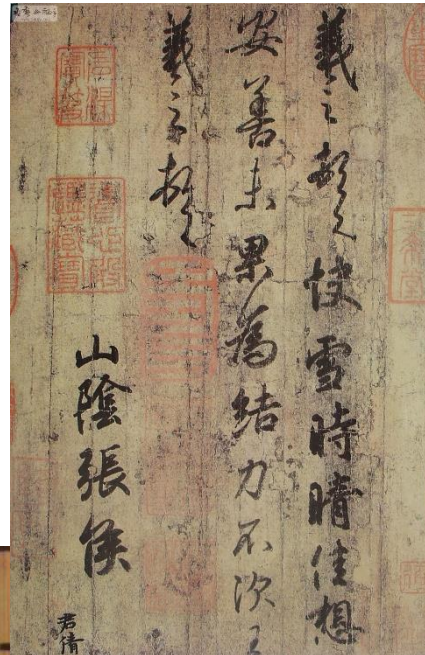
- Information Theory

Different Forms of Communications

■ Stone age: Shouting....



Ancient Time Communications

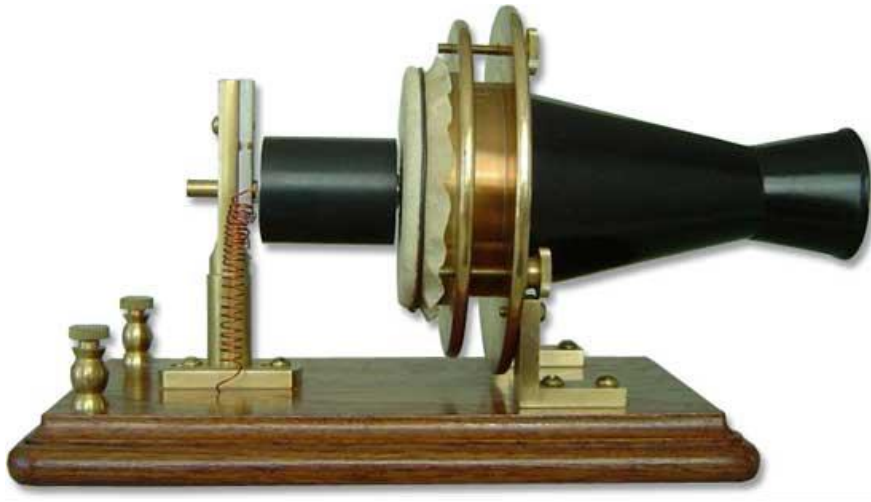


Letters



Beacon-fire

Bell Invented the Telephone



1876.03.10: “Mr. Watson – come here – I want to see you.”



1892 New York – Chicago Telephone line

Modern Communications: Phones



2G/GSM

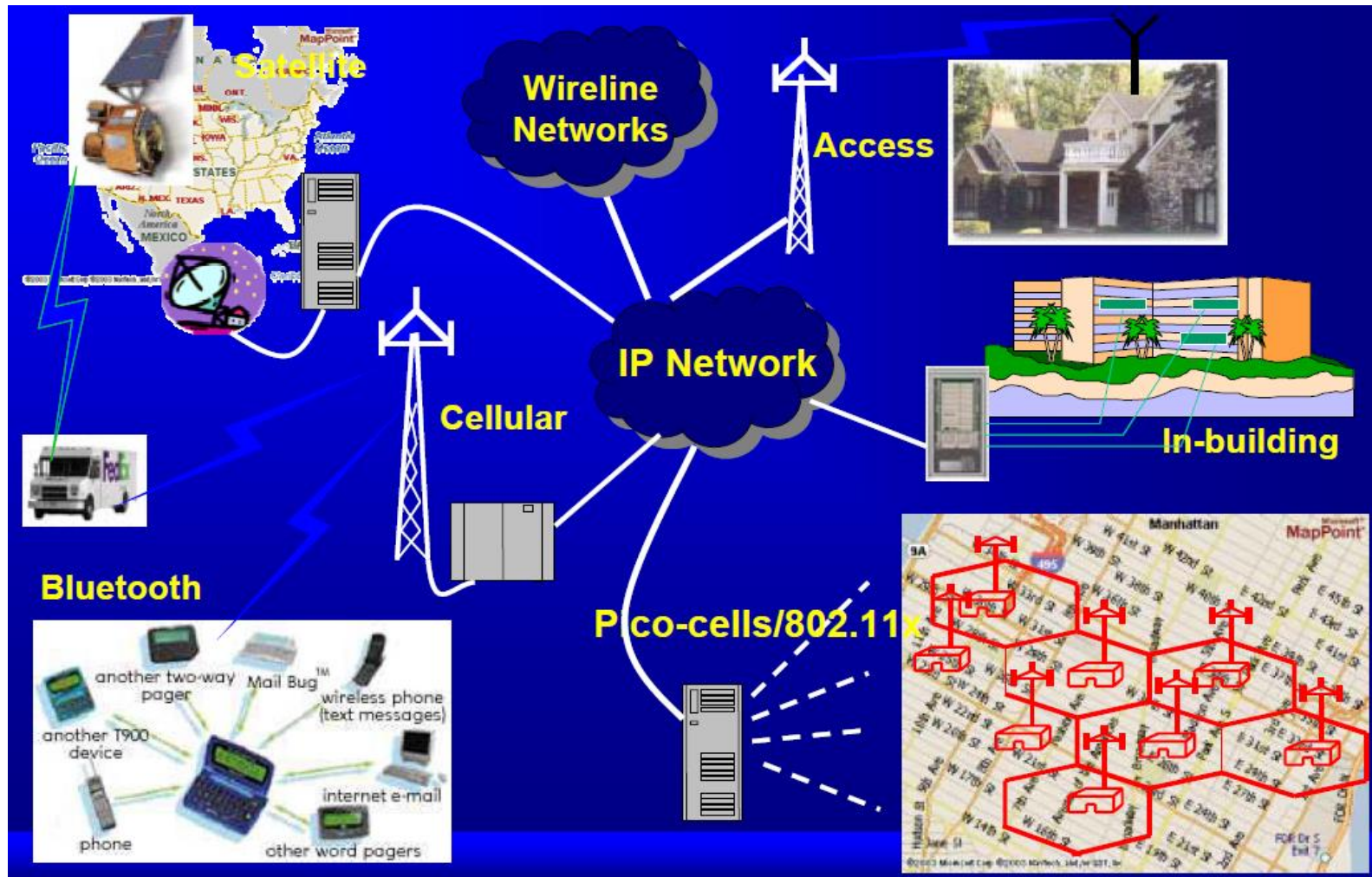


3G/4G

Wireline Phones

Cellular Phones

Modern Communications: Networks



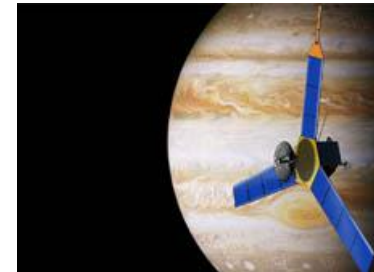
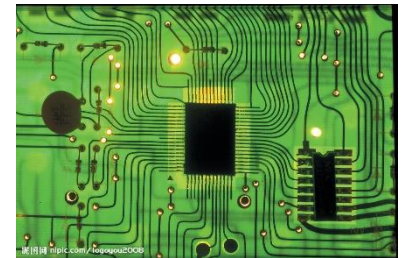
Communications Everywhere

■ Wireline

- ▶ Cables, hard discs, PCB board, chips

■ Wireless

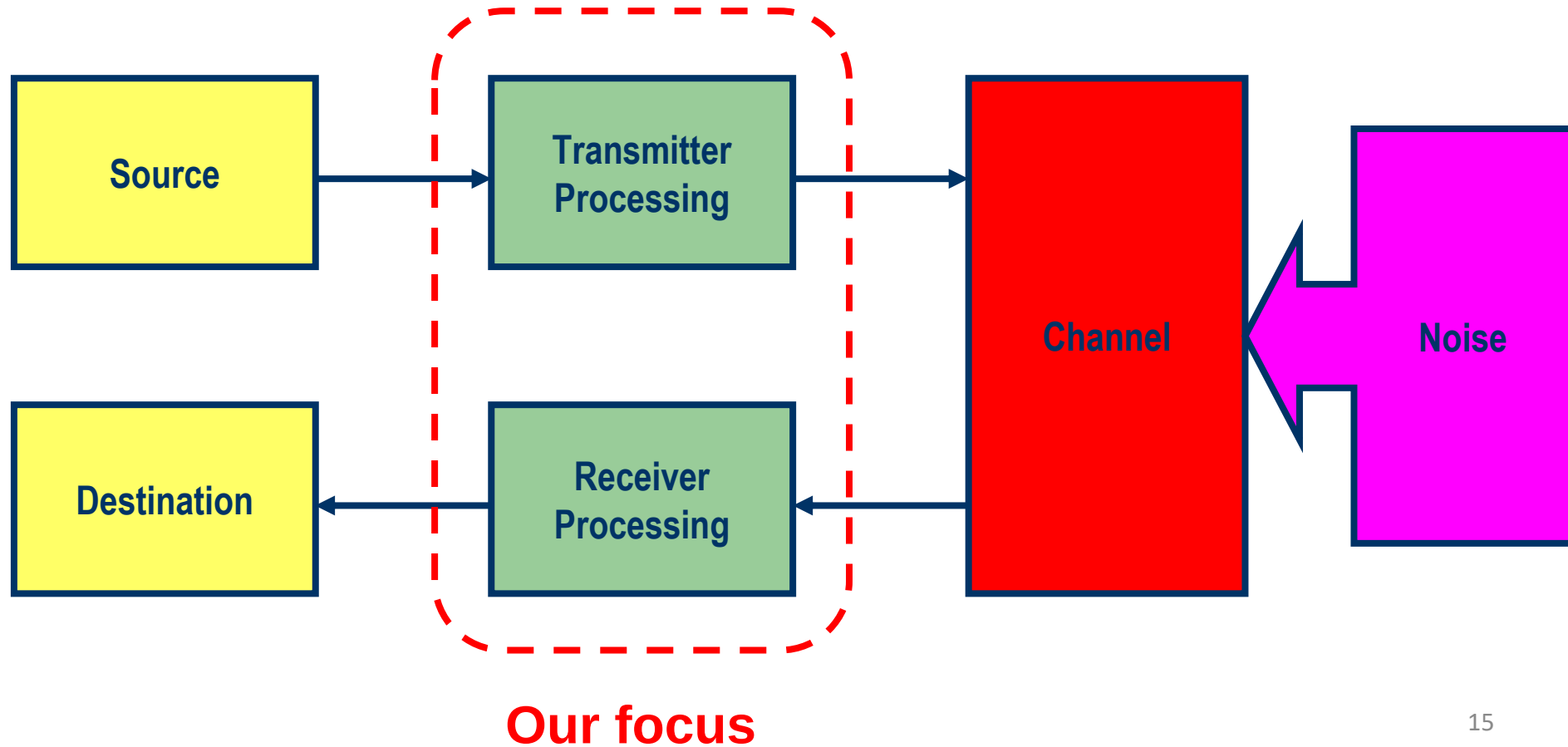
- ▶ Fixed radio, cellular communications, satellites



Communications: Concept

■ Communications:

- ▶ Carrying **information** over space and time, via **physical media**





Digital vs. Analog Communication

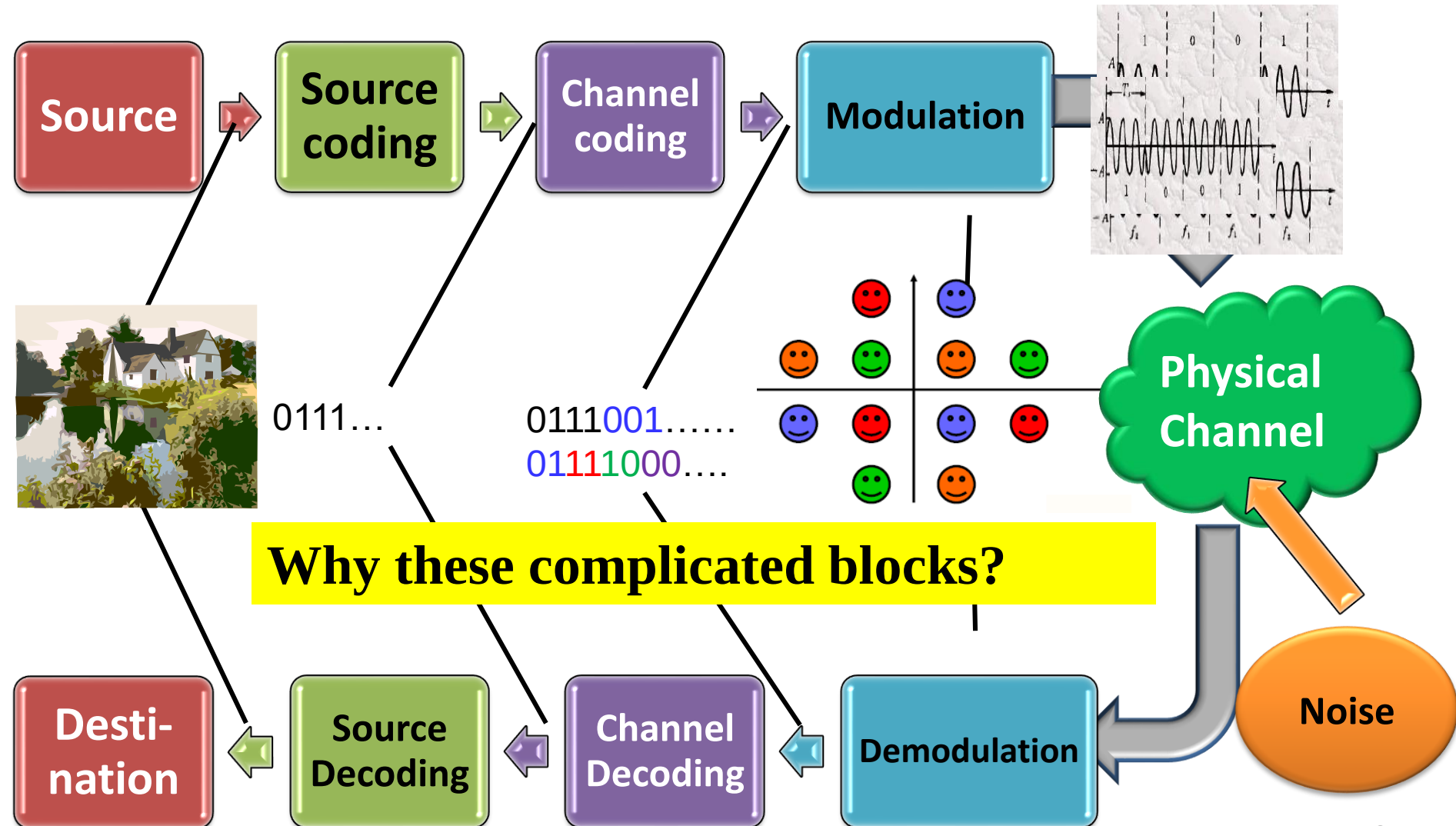
- Depending on the forms of the Tx/Rx processing
- Analog communications:
 - ▶ Communicating a **continuous-time waveform** (e.g., voltage from a microphone), via amplitude modulation (AM) or frequency modulation (FM), and etc.
- Digital communications (**our focus**):
 - ▶ Communicating a message comprising a **discrete-time sequence** of symbols from some **source alphabet** (generally **finite**)
 - ▶ Often mapped from a analog source (**source discretization**)
 - ▶ Often **coded into** a standard sequence of symbols, like **bits**
 - ▶ Often involving analog communication cross the physical channel



Digital or Analog Communications?

- Shouting
- Letter
- Beacon-fire
- Wireline telephone
- FM/AM radio
- Telegram with Morse code
- Cellular phone
- WIFI
- Main bus on the computer

Digital Communication Link: **Building Blocks**





Performance Metrics

■ Efficiency

- ▶ Better resource utilization: **less** Bandwidth (Hz)
- ▶ More information delivered: **high** Data rate (bits/second)

■ Reliability

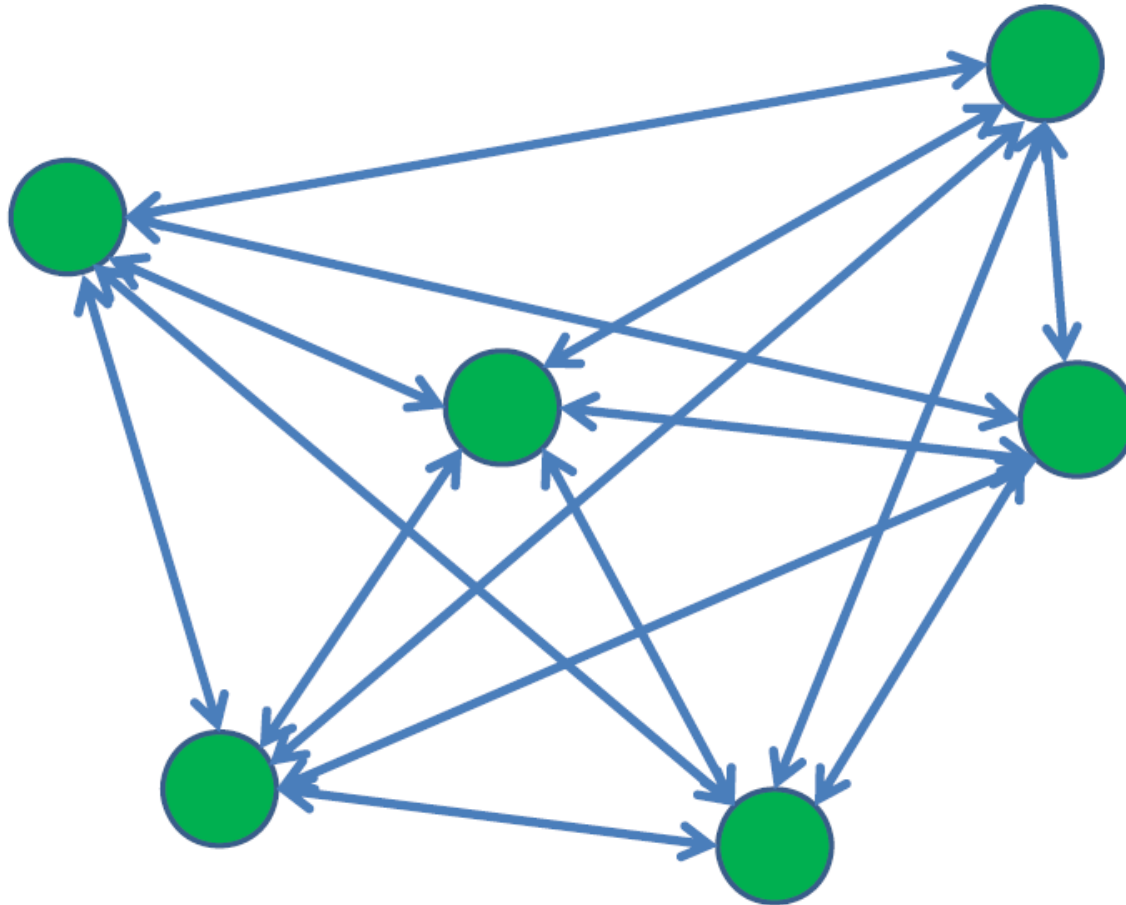
- ▶ Due to noise, error is inevitable
- ▶ **High** signal to noise ratio: **SNR** (dB)
- ▶ **Low** error probability
 - Bit error ratio (**BER**), for example: $P_b < 10^{-6}$



Beyond one link: Multiple Links

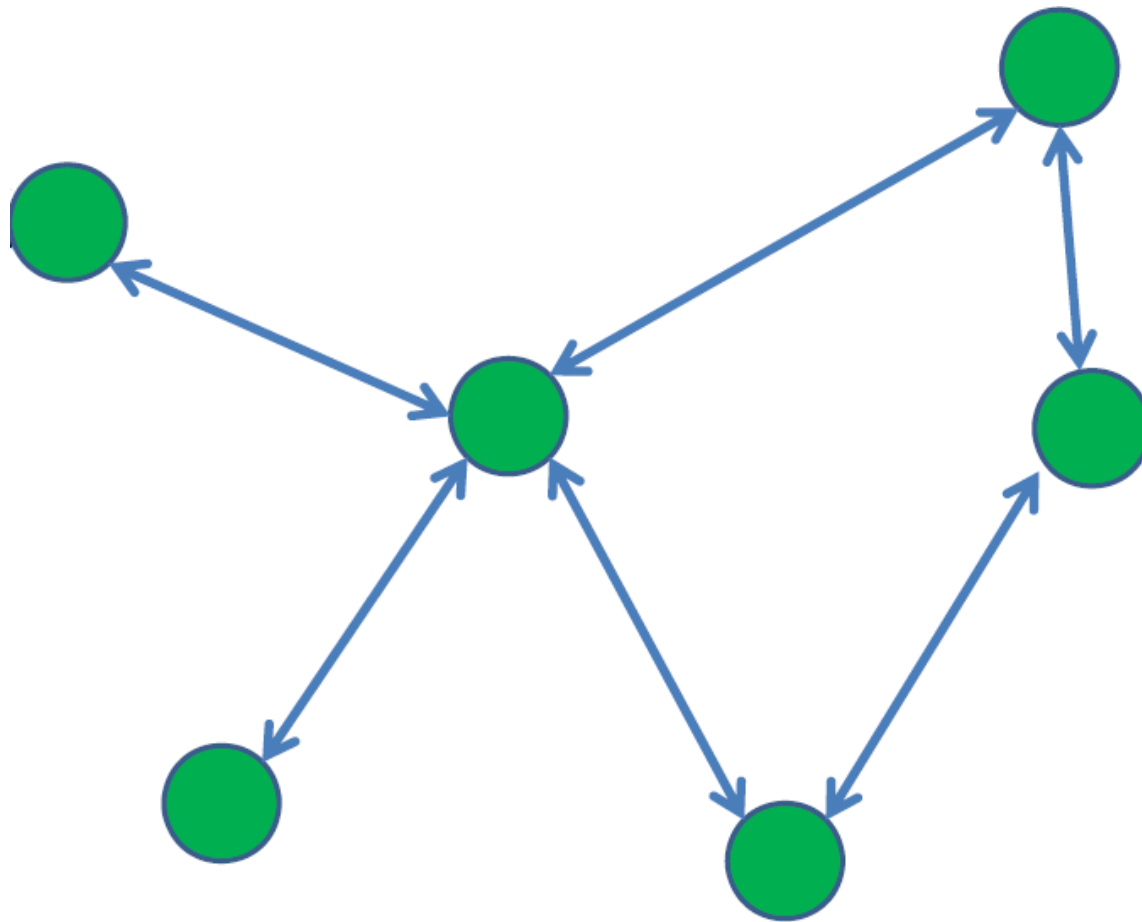
- Multiple links to communicate simultaneously
- Require resource sharing
- Time sharing:
 - ▶ Allocate different time: Time division multiple access (TDMA)
- Bandwidth sharing:
 - ▶ Allocate different frequencies: Frequency division multiple access (FDMA)
- Other types of resource sharing:
 - ▶ Code division multiple access (CDMA), Spatial division multiple access (SDMA),

Multi-point Communications



Communications among every pair of nodes

Multi-point Communications



...with intermediate nodes via switching and routing



Syllabus of the Course

■ Block by block, then to the network level

- ▶ Information theory and source coding (week 1 & 2)
- ▶ Source discretization (week 3)

Source Coding

- ▶ Baseband transmissions (week 4-6)
- ▶ Passband transmissions (week 7-9)

Modulation & Demodulation

- ▶ Channel coding (week 10 & 12)

- ▶ Multiplexing & multiple access (week 13)
- ▶ Switching & routing (week 14 & 15)
- ▶ TCP and transmission control (week 16)

Networks



Some Suggestions

- Understand each block clearly
 - ▶ Its input and output, design **goal**, key performance **metric** (**efficiency or reliability**)
 - ▶ The **theories** behind its **function**
 - Hints: review **Probability theory, Signals and systems, Linear algebra**
 - ▶ Revisit: Any alternative approach? Can we do better (**Optimality**)?

- Understand the interactions between adjacent blocks
 - ▶ How their input and output are connected
 - ▶ Revisit: Can they merge? Why are they still separated blocks? (**Layering**)



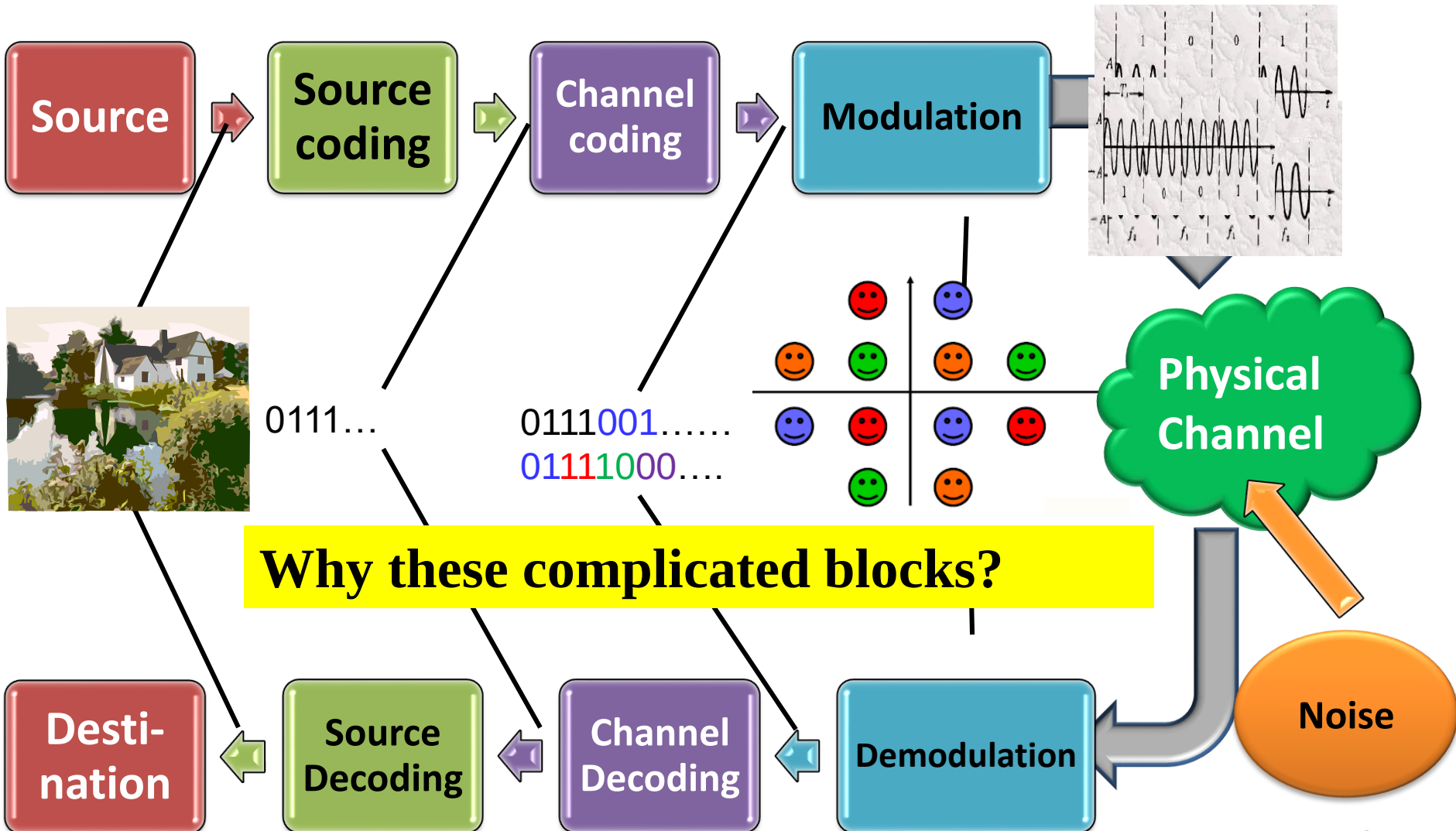
Outline

- Course Information
- Introduction
- Information Theory

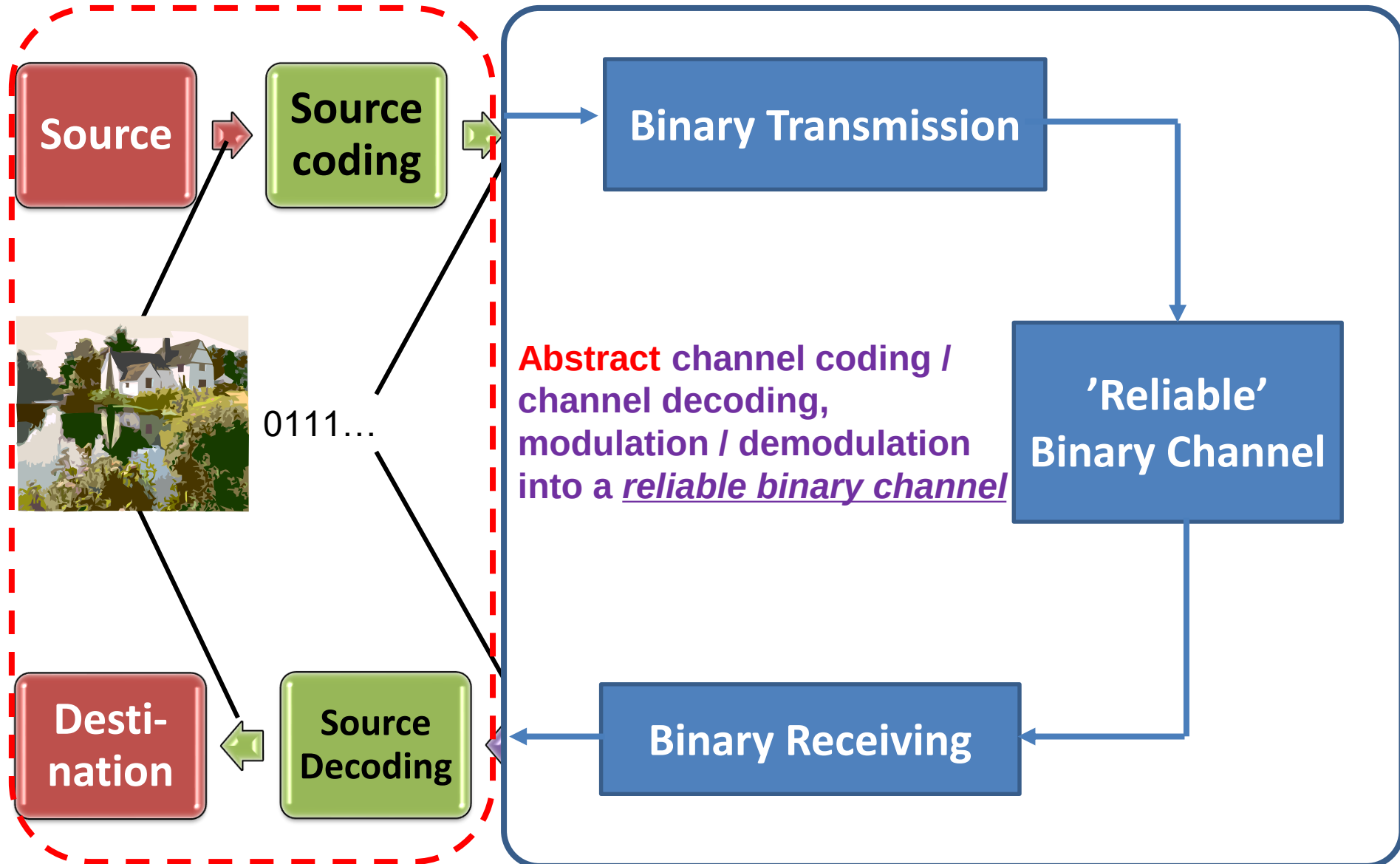


- From Entropy to Lossless Compression
- Joint Entropy and Conditional Entropy
- Mutual Information and Channel Capacity
- Differential Entropy, Mutual Information and Shannon's Formula

Start from Source Coding



Design Perspective of Source Coding



Source Coding: Overview

Next Chapter: Digitalization of Analog Source

Analog Input

Sampling

Quantization

Sample Sequence
(Discrete-Time)

Analog Output

Analog Filter

Reconstruction

This Chapter: Information Theory and Source Coding

Compression Encoder

Bit sequence
(0, 1)

'Reliable'
Binary Channel

Compression Decoder

Bit sequence
(0, 1)

Symbol Sequence
(Discrete-Time,
Discrete-Value)



The Bell System Technical Journal

Vol. XXVII

July, 1948

No. 3

A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist¹ and Hartley² on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have *meaning*; that is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem. The significant aspect is that the actual message is one *selected from a set* of possible messages. The system must be designed to operate for each possible selection, not just the one which will actually be chosen since this is unknown at the time of design.

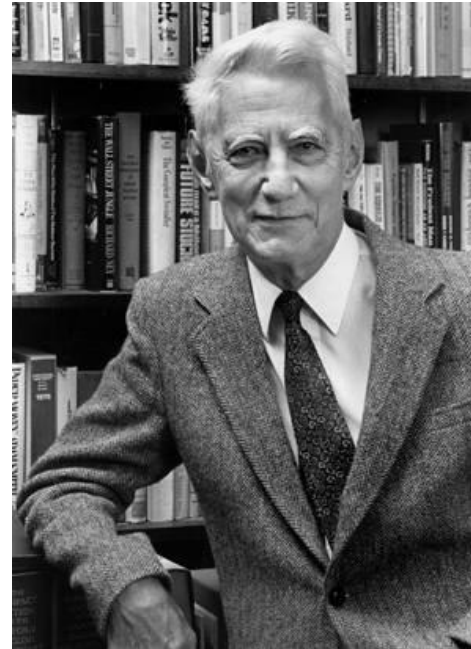
If the number of messages in the set is finite then this number or any monotonic function of this number can be regarded as a measure of the information produced when one message is chosen from the set, all choices being equally likely. As was pointed out by Hartley the most natural choice is the logarithmic function. Although this definition must be generalized considerably when we consider the influence of the statistics of the message and when we have a continuous range of messages, we will in all cases use an essentially logarithmic measure.

The logarithmic measure is more convenient for various reasons:

1. It is practically more useful. Parameters of engineering importance

¹ Nyquist, H., "Certain Factors Affecting Telegraph Speed," *Bell System Technical Journal*, April 1924, p. 324; "Certain Topics in Telegraph Transmission Theory," *A. I. E. E. Trans.*, v. 47, April 1928, p. 617.

² Hartley, R. V. L., "Transmission of Information," *Bell System Technical Journal*, July 1928, p. 535.



A Mathematical Theory of Communication

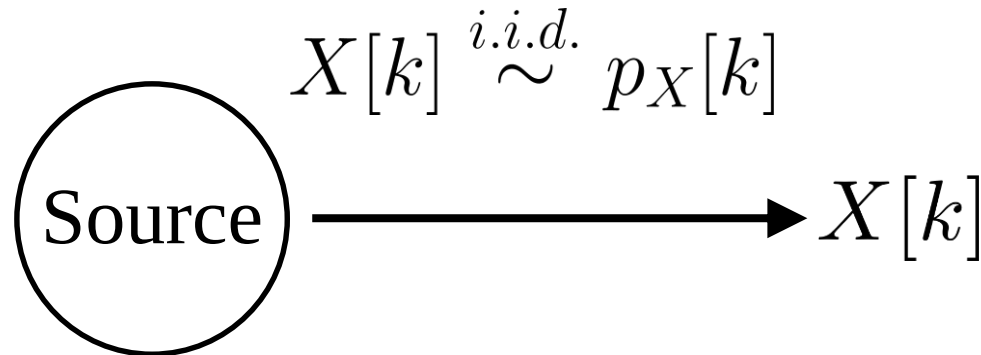


BSTJ, 1948

“The” Mathematical Theory of Communication

Entropy (1)

- Source: Generate independent and identically distributed (i.i.d) random variables (r.v.)



- Distribution of discrete random variables

$$X[k] \sim \begin{pmatrix} x_1 & x_2 & \cdots & x_N \\ p_1 & p_2 & \cdots & p_N \end{pmatrix}$$

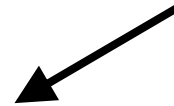
- Normalization

$$\sum_{k=1}^N p_k = 1, \quad p_k \geq 0$$

Entropy (2)

- Definition of Entropy:

$$\mathbb{E}_X(g(X))$$



$$H(X) = \mathbb{E}_X (-\log_2 \Pr \{X = x_i\})$$



$$H(X) = -\sum_{i=1}^N p_i \log p_i$$

Representation of Information

- **How to represent information in digital communications?**
- => **How to represent source symbols using bits?**
 - ▶ ASCII Table

十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符	十进制	字符		
0		16	►	32		48	0	64	@	80	P	96	`	112	p	128	Ç	144	É	160	á	176	▤	192	Ł	208	⌌	224	α	240	≡
1	☺	17	◄	33	!	49	1	65	A	81	Q	97	a	113	q	129	ü	145	æ	161	í	177	▥	193	┐	209	⌋	225	ß	241	±
2	☹	18	↕	34	"	50	2	66	B	82	R	98	b	114	r	130	é	146	Æ	162	ó	178	▦	194	└	210	⌌	226	Γ	242	≥
3	♥	19	!!	35	#	51	3	67	C	83	S	99	c	115	s	131	â	147	ô	163	ú	179	▧	195	┘	211	⌌	227	π	243	≤
4	♦	20	¶	36	\$	52	4	68	D	84	T	100	d	116	t	132	ä	148	ö	164	ñ	180	▨	196	—	212	⌋	228	Σ	244	∫
5	♣	21	§	37	%	53	5	69	E	85	U	101	e	117	u	133	à	149	ò	165	Ñ	181	▩	197	┘	213	⌋	229	σ	245	∫
6	♠	22	—	38	&	54	6	70	F	86	V	102	f	118	v	134	å	150	û	166	ª	182	▪	198	┘	214	⌋	230	μ	246	÷
7	•	23	↕	39	'	55	7	71	G	87	W	103	g	119	w	135	ç	151	ù	167	º	183	▫	199	┘	215	⌋	231	τ	247	≈
8	▣	24	↑	40	(	56	8	72	H	88	X	104	h	120	x	136	ê	152	ÿ	168	¿	184	▬	200	┘	216	⌋	232	Φ	248	°
9	○	25	↓	41	)	57	9	73	I	89	Y	105	i	121	y	137	ë	153	Ö	169	¬	185	▮	201	┘	217	⌋	233	Θ	249	•
10	◉	26	→	42	*	58	:	74	J	90	Z	106	j	122	z	138	è	154	Ü	170	¬	186	▯	202	⌌	218	┘	234	Ω	250	•
11	♂	27	←	43	+	59	;	75	K	91	[	107	k	123	{	139	ï	155	☹	171	½	187	▰	203	┘	219	▣	235	δ	251	√
12	♀	28	└	44	,	60	<	76	L	92	\	108	l	124		140	î	156	£	172	¼	188	▱	204	┘	220	▣	236	∞	252	n
13	♪	29	↔	45	-	61	=	77	M	93	]	109	m	125	}	141	ì	157	¥	173	;	189	▲	205	=	221	▣	237	φ	253	²
14	🎵	30	▲	46	.	62	>	78	N	94	^	110	n	126	~	142	Ä	158	ℙ	174	«	190	△	206	┘	222	▣	238	∈	254	■
15	☼	31	▼	47	/	63	?	79	O	95	_	111	o	127	◊	143	Â	159	f	175	»	191	▴	207	⌌	223	▣	239	∩	255	ÿ



Source Coding: Compression Coding

Next Chapter: Digitalization
of Analog Source

Analog
Input

Sampling

Quantization

Sampled
Sequence
(Discrete-Time)

Analog
Output

Analog
Filter

Reconstruction

This Chapter: Information
Theory and Source Coding

Compression
Encoder

Bit sequence
(0, 1)

'Reliable'
Binary Channel

Compression
Decoder

Bit sequence
(0, 1)

Symbol
Sequence
(Discrete-Time,
Discrete-Value)

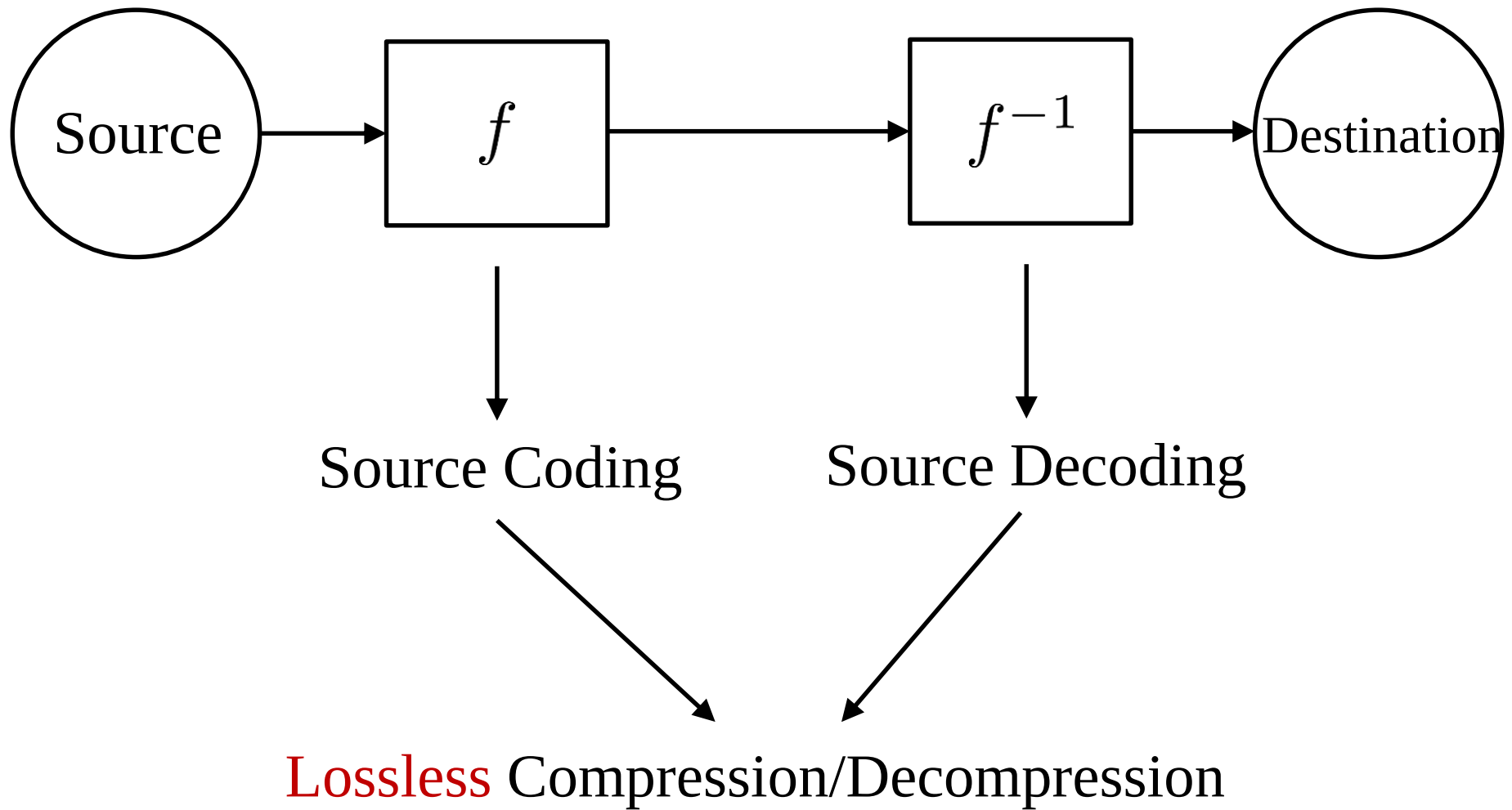


Average Codeword Length (1)

- How many bits in average to represent a discrete-value random variable from the source?
- Design a function f to map $X[k]$ into a bit sequence (codeword)

$$f : X[1], X[2], \dots, X[k] \mapsto \underbrace{01001 \dots 110}$$

How many bits?





Average Codeword Length (2)

- For source coding

$$f : X[1], X[2], \dots, X[k] \mapsto \underbrace{010 \dots 1}_{\text{Coded sequence}}$$

- If $\exists f^{-1}$, then the least average codeword length

$$\approx kH(X)$$

Average Codeword Length (3)

- Define $l : \underbrace{010 \dots 1}_L \mapsto \boxed{L}$ Number of bits in a sequence

Then

$$\frac{1}{K} \sum_{k=1}^K l(f(X[k])) \xrightarrow{K \rightarrow \infty} H(X)$$

- Because $X[k]$ is i.i.d., we know $l(f(X[k]))$ is ergodic
- Therefore,

$$H(X) = \mathbb{E}_X (l(f(X)))$$



What is information?

- Information to reduce uncertainty

- ▶ In Information Theory, the uncertainty is due to randomness
- ▶ Thus we describe things using random variables

- Example

- ▶ Flip a coin: '1' for head, '0' for tail
- ▶ We know the prior distribution

$$X \sim \begin{pmatrix} 0 & 1 \\ 0.5 & 0.5 \end{pmatrix}$$

- ▶ After observing the result, we eliminate the uncertainty of a random event, or obtain the information!

“Information is the resolution of uncertainty” - C.E. Shannon



The measure of information

- How to compare the amount of information
 - ▶ Intuitively,
 - ▶ In a lottery setting:
 - ▶ No prize —— common, less information
 - ▶ Top Prize —— super rare, more information

In Chinese, “以稀为贵”

More information with rare events (with small probability)



Amount of information for events (1)

- Define the amount of information obtained by observing event $X = x_k$:

$$H(X = x_k) = \log \left(\frac{1}{p_k} \right)$$

- Property 1: A event in certain has zero amount of information

$$H(X = x_k) = 0 \quad \text{for } p_k = 1$$

- Property 2: Every event has non-negative amount of information

$$H(X = x_k) \geq 0 \quad \text{for } 0 \leq p_k \leq 1$$



Amount of information for events (2)

- Property 3: Rarer event has more information

$$H(X = x_k) > H(X = x_i) \quad \text{for } p_k < p_i$$

- Property 4: (Additivity)

If x_k, x_l are statistically independent, then

$$H(x_k, x_l) = H(X = x_k) + H(X = x_l)$$

- Unit:
- base 2: bit
- base e : nat



Amount of information for events (3)

- Let $H(X = x_i) = f(p_i)$
- Flip the coin twice. If both are head, i.e. $\{1,1\}$, then the information obtained can be calculated as:

► Method 1: Observe once after all

$$f\left(\frac{1}{2} \times \frac{1}{2}\right) = f\left(\frac{1}{4}\right)$$

► Method 2: Successive observations

$$f\left(\frac{1}{2}\right) + f\left(\frac{1}{2}\right)$$

Note: Methods of observation do not change the amount of information obtained



Amount of information for events (4)

- Thus, $H(X = x_i) = f(p_i)$ should satisfy:

$$f\left(\frac{1}{4}\right) = f\left(\frac{1}{2}\right) + f\left(\frac{1}{2}\right)$$

- Meanwhile, the amount of information is non-negative
- Therefore,

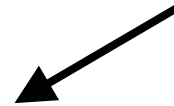
$$H(X = x_i) = -\log p_i$$

This only gives the amount of information for a particular result



Recap the definition of entropy

$$\mathbb{E}_X(g(X))$$



$$H(X) = \mathbb{E}_X (-\log_2 \Pr \{X = x_i\})$$



$$H(X) = -\sum_{i=1}^N p_i \log p_i$$

The entropy of X equals to the average information of all possible events



Entropy and Information (1)

- A proper measure of information should have following properties:
- **Continuity:** Entropy should be continuous to the probabilities of events
- **Additivity:** If the events are grouped, and the experiment is decomposed to two successive experiments according to the groups, then the entropy of original experiment equals to the entropy of the **first experiment on groups** plus the **probability-weighted sum of the entropy of second experiments**
- **Monotonicity** ——For equiprobable events, the entropy should increase with the number of events
- **Theorem: The only form of entropy satisfying above properties is**

$$H(X) = - \sum_{i=1}^N p_i \log p_i$$



Entropy and Information (2)

■ Proof:

- Denote the entropy of the experiment with n equiprobable events by $f(n)$. Due to monotonicity, $f(n)$ is monotonically increasing. Note that the experiment with mn equiprobable events can be decomposed into m experiments with n equiprobable events. By additivity,

$$\begin{aligned} f(mn) &= f(m) + \underbrace{\frac{1}{m}f(n) + \cdots + \frac{1}{m}f(n)}_m \\ &= f(m) + f(n) \end{aligned}$$



Entropy and Information (3)

- Proof (Cont'd):
- The monotonic function satisfying $f(mn) = f(m) + f(n)$ must have the form

$$f(n) = \log(n)$$

- Further, consider a general random variable

$$X \sim \begin{pmatrix} x_1 & x_2 & \cdots & x_N \\ p_1 & p_2 & \cdots & p_N \end{pmatrix}$$

where p_i are rational numbers, thus

$$p_i = \frac{k_i}{\sum_{i=1}^n k_i}$$



Entropy and Information (4)

- Proof (Cont'd): Consider a experiment with

$$\sum_{i=1}^N k_i$$

equiprobable events. It can be decomposed into two successive experiments. The former generates x_i with probability p_i . The latter experiment conditioned on x_i generates k_i equiprobable experiments. By additivity,

$$\log \sum_{i=1}^N k_i = H(X) + \sum_{i=1}^N p_i \log k_i$$



Entropy and Information (5)

■ Proof (Cont'd): By

$$\log \sum_{i=1}^N k_i = H(X) + \sum_{i=1}^N p_i \log k_i$$

■ Then

$$H(X) = \log \sum_{i=1}^N k_i - \sum_{i=1}^n p_i \log k_i$$

$$= \sum_{i=1}^N p_i \log \sum_{i=1}^n k_i - \sum_{i=1}^n p_i \log k_i$$

$$= \sum_{i=1}^N p_i \log \frac{\sum_{i=1}^n k_i}{k_i}$$

$$= \sum_{i=1}^N p_i \log \frac{1}{p_i}$$



Entropy and Information (5)

■ For

$$X \sim \begin{pmatrix} x_1 & x_2 & \cdots & x_N \\ p_1 & p_2 & \cdots & p_N \end{pmatrix}$$

where p_i is a real number.

- Note that any real number can be approximated by a rational number with arbitrary precision
- Therefore, the entropy of any discrete random variable expressed as:

$$H(X) = - \sum_{i=1}^N p_i \log p_i$$



Properties of Entropy (1)

- The entropy of X as defined as

$$H(X) = \sum_{k=0}^{K-1} p_k \log_2 \left(\frac{1}{p_k} \right)$$

where K is the number of symbols (events). **We have**

$$0 \leq H(X) \leq \log_2 K$$

- $H(X) = 0$ if and only if $\exists k, p_k = 1$ (i.e. certain)

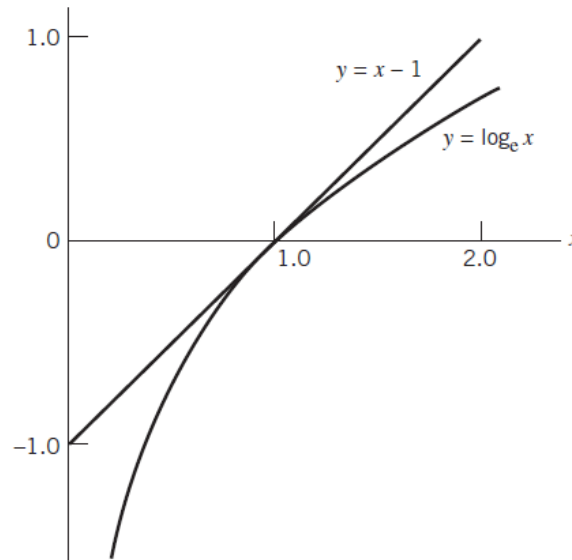
- Proof:

By definition, $H(X) \geq 0$, verify the example above with zero entropy.



Properties of Entropy (2)

- $H(X) = \log K$ if and only if the symbols from the source are equiprobable
- The upper bound of entropy corresponds to the maximum uncertainty
 - ▶ Proof:
 - ▶ Using inequality $\log_e x \leq x - 1, \quad x \geq 0$





Properties of Entropy (3)

- ▶ Let $p_0, p_1, \dots, p_{K-1}, q_0, q_1, \dots, q_{K-1}$ be two probability distributions on an alphabet

$$\mathcal{X} = \{x_0, x_1, \dots, x_{K-1}\}$$

- ▶ Define (**Relative Entropy**)

$$D(p||q) = \sum_{k=0}^{K-1} p_k \log_2 \left(\frac{p_k}{q_k} \right)$$



Properties of Entropy (4)

► By inequality $\log_e x \leq x - 1$, $x \geq 0$
we have

$$\begin{aligned}\sum_{k=0}^{K-1} p_k \log_2 \left(\frac{p_k}{q_k} \right) &= - \sum_{k=0}^{K-1} p_k \log_2 \left(\frac{q_k}{p_k} \right) \\ &\geq - \frac{1}{\ln 2} \sum_{k=0}^{K-1} p_k \left(\frac{q_k}{p_k} - 1 \right) \\ &= - \frac{1}{\log_e 2} \sum_{k=0}^{K-1} (q_k - p_k) \\ &= 0\end{aligned}$$



Properties of Entropy (5)

- ▶ Therefore, $D(p||q) \geq 0$ and the equality holds if and only if p and q have the same distribution
- ▶ Now we prove the upper bound of entropy using the non-negative property of D :
- ▶ Let q be the uniform distribution on $\mathcal{X}$, i.e.

$$q_k = \frac{1}{K}, \quad k = 0, 1, \dots, K-1$$

- ▶ For any distribution p , we have

$$\begin{aligned} D(p||q) &= \sum_{k=0}^{K-1} p_k \log_2 p_k + \log_2 K \sum_{k=0}^{K-1} p_k \\ &= -H(X) + \log_2 K \end{aligned}$$



Properties of Entropy (6)

► Thus,

$$\begin{aligned} D(p||q) &= \sum_{k=0}^{K-1} p_k \log_2 p_k + \log_2 K \sum_{k=0}^{K-1} p_k \\ &= -H(X) + \log_2 K \end{aligned}$$

► Because D is non-negative,

$$H(X) \leq \log_2 K$$

if and only if the symbols are equiprobable. ■

Entropy of Discrete R.V.: Examples

- Entropy of a coin? Entropy of lottery?
- Entropy of beacon-fire?





Entropy of a Bernoulli R.V. (1)

- Consider a Bernoulli source. Symbol ‘0’ has probability p_0 , while symbol ‘1’ has probability $1 - p_0$

$$\begin{aligned} H(X) &= -p_0 \log_2 p_0 - p_1 \log_2 p_1 \\ &= -p_0 \log_2 p_0 - (1 - p_0) \log_2 (1 - p_0) \text{ bits} \end{aligned}$$

- Properties:

- ▶ Symmetry: The entropy is symmetric to $p_0 = \frac{1}{2}$

- ▶ Minimum value:

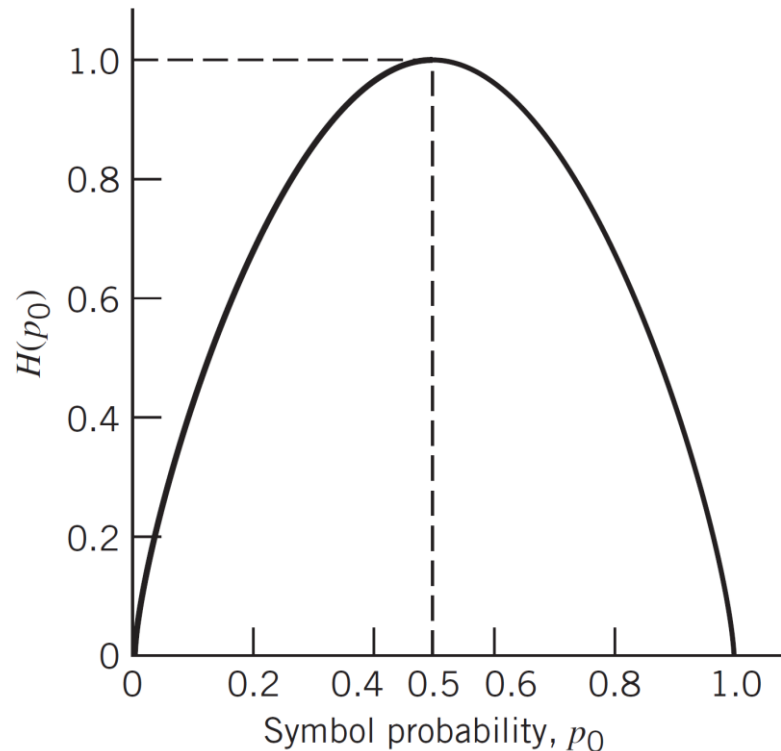
$$p_0 = 1, H(X) = 0 \quad p_0 = 0, H(X) = 0$$

- ▶ Maximum value at $p_0 = \frac{1}{2}$, $H_{\max} = 1$

Entropy of Bernoulli R.V. (2)

- Define the entropy function

$$H(p_0) = -p_0 \log_2 p_0 - (1 - p_0) \log_2 (1 - p_0) \text{ bits}$$





An Example (1)

- Let X be a discrete r.v. with distribution

$$X \sim \left(\begin{array}{cccc} 0, & 1, & 2, & 3 \\ \frac{1}{2}, & \frac{1}{4}, & \frac{1}{8}, & \frac{1}{8} \end{array} \right).$$

- Calculate the entropy of X

$$H(X) = ?$$



An Example (2)

- By definition,

$$\begin{aligned} H(X) &= \frac{1}{2} \log 2 + \frac{1}{4} \log 4 + \left(\frac{1}{8} \log 8 \right) \times 2 \\ &= 1\frac{3}{4}. \end{aligned}$$

- Now we look for an encoding function f satisfying:

$$\begin{cases} \textcircled{1} & \exists f^{-1}, \\ \textcircled{2} & \mathbb{E}_X (l(f(X))) = 1\frac{3}{4}. \end{cases}$$

An Example (3)

$$X \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \end{pmatrix}$$

- One example: (although not a good enough code)

$$f(0) = 00 \quad f(1) = 01 \quad f(2) = 10 \quad f(3) = 11$$

- $\exists f^{-1}$



- $\mathbb{E}_X (l(f(X))) = 2 > \frac{7}{4}$



An Example (4)

$$X \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \end{pmatrix}$$

- A better code (mapping)

$$f(0) = 0 \quad f(1) = 10 \quad f(2) = 110 \quad f(3) = 111$$

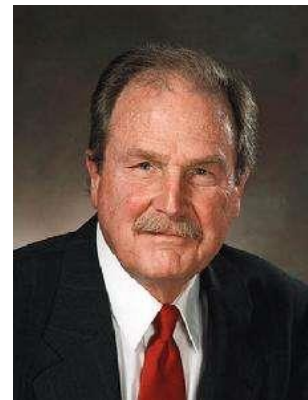
- $\exists f^{-1}$



- $\mathbb{E}_X (l(f(X))) = \frac{7}{4}$

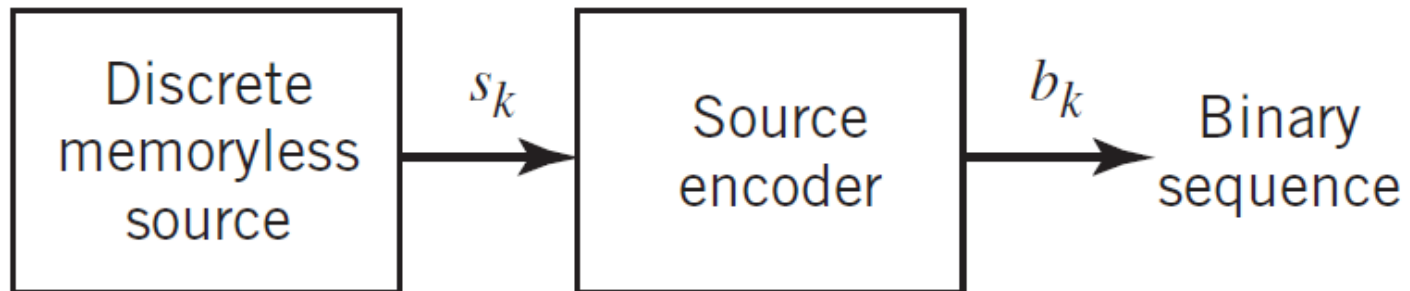


**Huffman
Code**



Lossless Source Compression (1)

- How to represent the data generated from discrete source?
 - ▶ Two fundamental requirements on source coding:
 1. The codewords produced by the encoder are in binary form
 2. The code is **uniquely decodable**, so that the original source sequence can be reconstructed perfectly from the encoded binary sequence





Lossless Source Compression (2)

- Symbol s_k has probability p_k
- Symbol s_k corresponds to codeword b_k , with length l_k
- The average codeword length is

$$\bar{L} = \sum_{k=0}^{K-1} p_k l_k$$

- The lower bound of average codeword length (Shannon's first theorem)

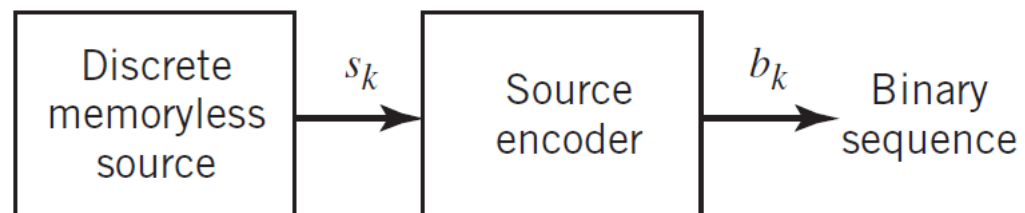
$$\bar{L} \geq H(S)$$

- Coding Efficiency ($\eta \leq 1$) $\eta = \frac{H(S)}{\bar{L}}$



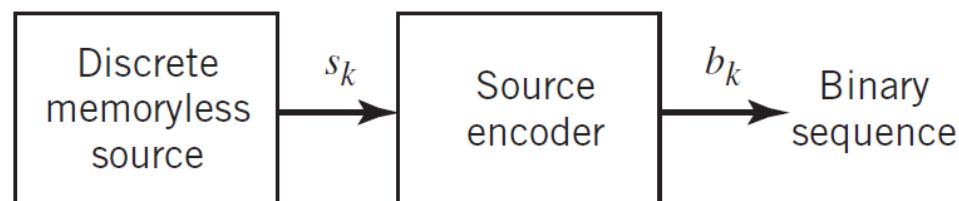
Lossless Data Compression Algorithms

- In fact, the symbols in the source contain many redundancies
- Target: eliminate redundancy and increase the efficiency of transmission and storage
- Lossless compression: **reduce redundancy** and can still reconstruct **without information loss**
 - Shorter codewords are assigned to symbols which appear more frequently
 - ✓ The lower bound of the average codeword length of lossless compression is given by **the entropy of the source**



Prefix Coding (1)

- The source has alphabet $\{s_0, s_1, \dots, s_{K-1}\}$ with probabilities $\{p_0, p_1, \dots, p_{K-1}\}$
- The binary codeword for symbol s_k is $(m_{k_1}, m_{k_2}, \dots, m_{k_n})$
- The prefix of a codeword is the initial i digits $(m_{k_1}, \dots, m_{k_i})$ for some $i \leq n$
- Then the prefix coding is defined as (**WHY?**):
 - ▶ No codeword is the prefix of any other codeword





Prefix Coding (2)

- Question: Which code is prefix code?
- Which is uniquely decodable?

Symbol source	Probability of occurrence	Code I	Code II	Code III
s_0	0.5	0	0	0
s_1	0.25	1	10	01
s_2	0.125	00	110	011
s_3	0.125	11	111	0111

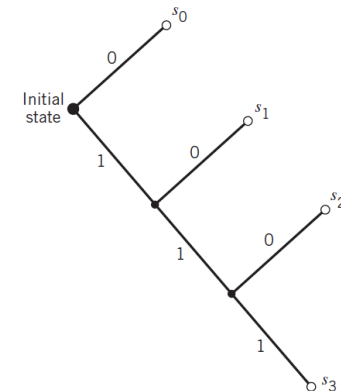
Prefix Coding (3)

- Kraft's Inequality: necessary condition for a uniquely decodable code

$$\sum_{k=0}^{K-1} 2^{-l_k} \leq 1$$

- Code I does not satisfies Kraft's inequality
- Code II & Code III satisfies Kraft's inequality
- But only Code II is a prefix code

Symbol source	Probability of occurrence	Code I	Code II	Code III
s_0	0.5	0	0	0
s_1	0.25	1	10	01
s_2	0.125	00	110	011
s_3	0.125	11	111	0111



Prefix Coding (4)

■ Intuition on Kraft's Inequality

$$\sum_{x \in \mathcal{X}} 2^{-l(x)} \leq 1$$

Length of segments

Turn each codeword to a binary decimal, i.e.

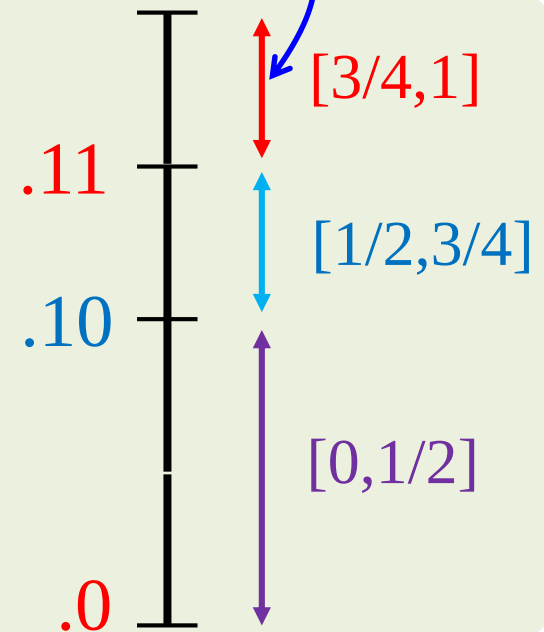
codeword $y_1 y_2 \dots y_l$

to $y_1/2 + y_2/4 + \dots + y_l 2^{-l}$

Example: .011 to $1/4 + 1/8$

Then each codeword corresponds to a segment

$$[\sum_{i=1}^l y_i 2^{-i}, \sum_{i=1}^l y_i 2^{-i} + 2^{-l})$$



Prefix coding $\rightarrow$ Non overlap of the segments



Prefix Coding (5)

- We can construct a prefix code which satisfies

$$H(S) \leq \bar{L} < H(S) + 1$$

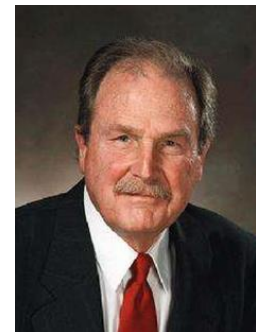
- The left-hand-side is satisfied with equality when the probabilities of source symbols satisfies

$$p_k = 2^{-l_k}$$

where l_k is the codeword length of symbol s_k

Huffman Coding (1)

1. The source symbols are listed in order of decreasing probability. The two source symbols of lowest probability are assigned 0 and 1.
2. These two source symbols are then combined into a new source symbol with probability equal to the sum of the two original probabilities
3. The procedure is repeated until we are left with only two for which the symbols 0 and 1 are assigned.





Huffman Coding (2)

- Remark: Huffman code is the optimal prefix code for discrete memoryless source, in the sense of shortest expected length.
- It also satisfies $H(S) \leq \bar{L} < H(S) + 1$

$$H(S) = 2.121 \text{ bits}$$

- An example:

<u>Symbol</u>	<u>Probability</u>
s_0	0.4
s_1	0.2
s_2	0.2
s_3	0.1
s_4	0.1

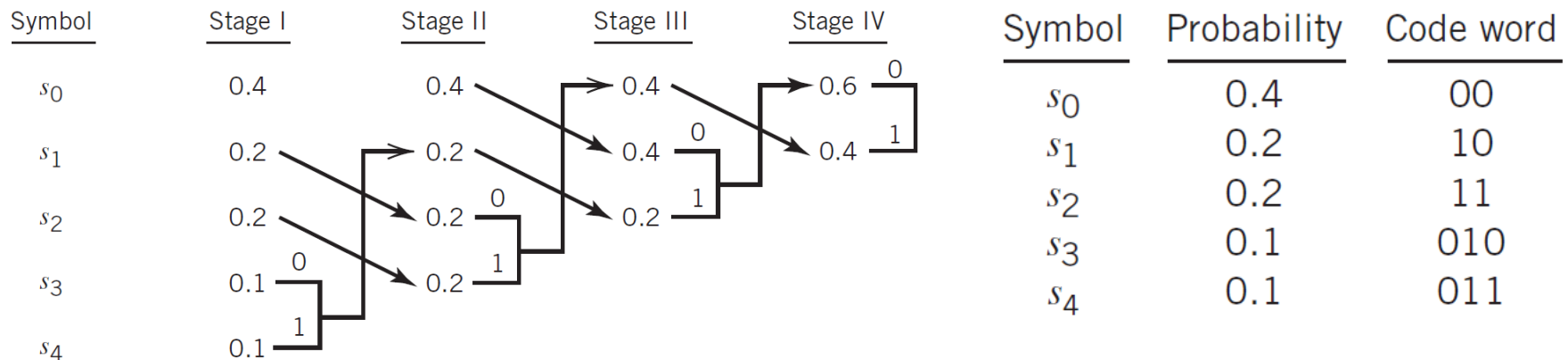


Huffman Coding (3)

■ An example: $H(S) \leq \bar{L} < H(S) + 1$

$$H(S) = 2.121 \text{ bits}$$

$$\bar{L} = 2.2 \text{ binary symbols}$$



■ Two observations on this example:

1. The coding efficiency is 96.4%
2. The codeword length satisfies Kraft's inequality



- From Entropy to Lossless Compression
- Joint Entropy and Conditional Entropy
- Mutual Information and Channel Capacity
- Differential Entropy, Mutual Information and Shannon's Formula



Joint Entropy of two Random Variables

- Consider two random variables

$$XY \sim p_{ij} = \Pr \{X = x_i, Y = y_j\}$$


Joint Distribution

- Q: How to define the joint entropy of X and Y?



Joint Entropy of two Random Variables

$$\begin{aligned} H(XY) &= \mathbb{E}_{XY} (-\log_2 \Pr \{X = x_i, Y = y_j\}) \\ &= - \sum_{i=1}^N \sum_{j=1}^M p_{ij} \log_2 p_{ij} \end{aligned}$$

- Measure of uncertainty in r.v. pairs (X, Y)



Conditional Entropy (1)

$$X \mid Y \sim p_{i|j} = \frac{\Pr \{X = x_i, Y = y_j\}}{\Pr \{Y = y_j\}}$$

↓

Conditional Distribution

$$Y \mid X \sim p_{j|i}$$

$$p_{ij} = p_j p_{i|j} = p_i p_{j|i}$$



Conditional Entropy (2)

$$\begin{aligned} H(X \mid Y) &= \mathbb{E}_{XY} (-\log_2 \Pr \{X = x_i \mid Y = y_j\}) \\ &= - \sum_{i=1}^N \sum_{j=1}^M \underline{p_{ij}} \log (\underline{p_{i|j}}) \end{aligned}$$

$$H(Y \mid X) = - \sum_{i=1}^N \sum_{j=1}^M \underline{p_{ij}} \log (\underline{p_{j|i}})$$

- Q: What is the meaning of conditional entropy?



Chain Rule (1)

$$\begin{aligned} H(X | Y) &= - \sum_{i=1}^N \sum_{j=1}^M p_{ij} \log \left(\frac{p_{ij}}{p_j} \right) \\ &= - \sum_{i=1}^N \sum_{j=1}^M p_{ij} (\log(p_{ij}) - \log(p_j)) \\ &= - \sum_{i=1}^N \sum_{j=1}^M p_{ij} \log(p_{ij}) - \left(- \sum_{j=1}^M \left(\sum_{i=1}^N p_{ij} \right) \log(p_j) \right) \\ &= H(XY) - \left(- \sum_{j=1}^M p_j \log(p_j) \right) \\ &= H(XY) - H(Y) \end{aligned}$$



Chain Rule (2)

$$H(X|Y) = H(XY) - H(Y).$$

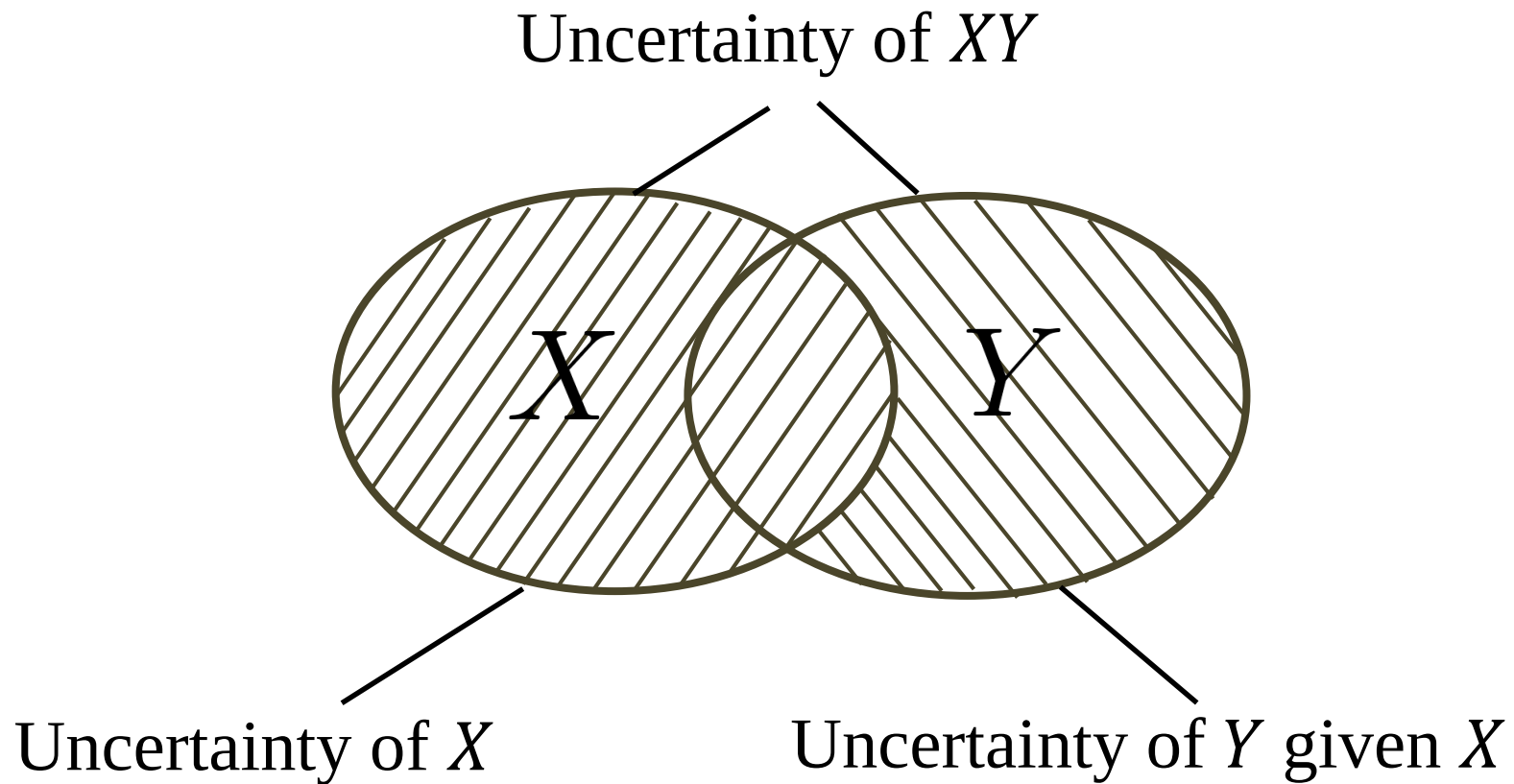
- Uncertainty of X given Y

$$H(XY) = H(Y) + H(X|Y).$$

$$H(XY) = H(X) + H(Y|X).$$

Chain Rule (3)

$$H(XY) = H(X) + H(Y|X).$$



Chain Rule (4)

- For independent random variables $X \perp Y$

$$p_{ij} = p_i p_j$$

$$\iff H(XY) = H(X) + H(Y)$$

$$\iff H(X|Y) = H(X)$$

- The information of Y does not help in reducing uncertainty of X
- Additivity of entropy



Chain Rule (5)

- Random variable $Y = f(X)$ is a function of X

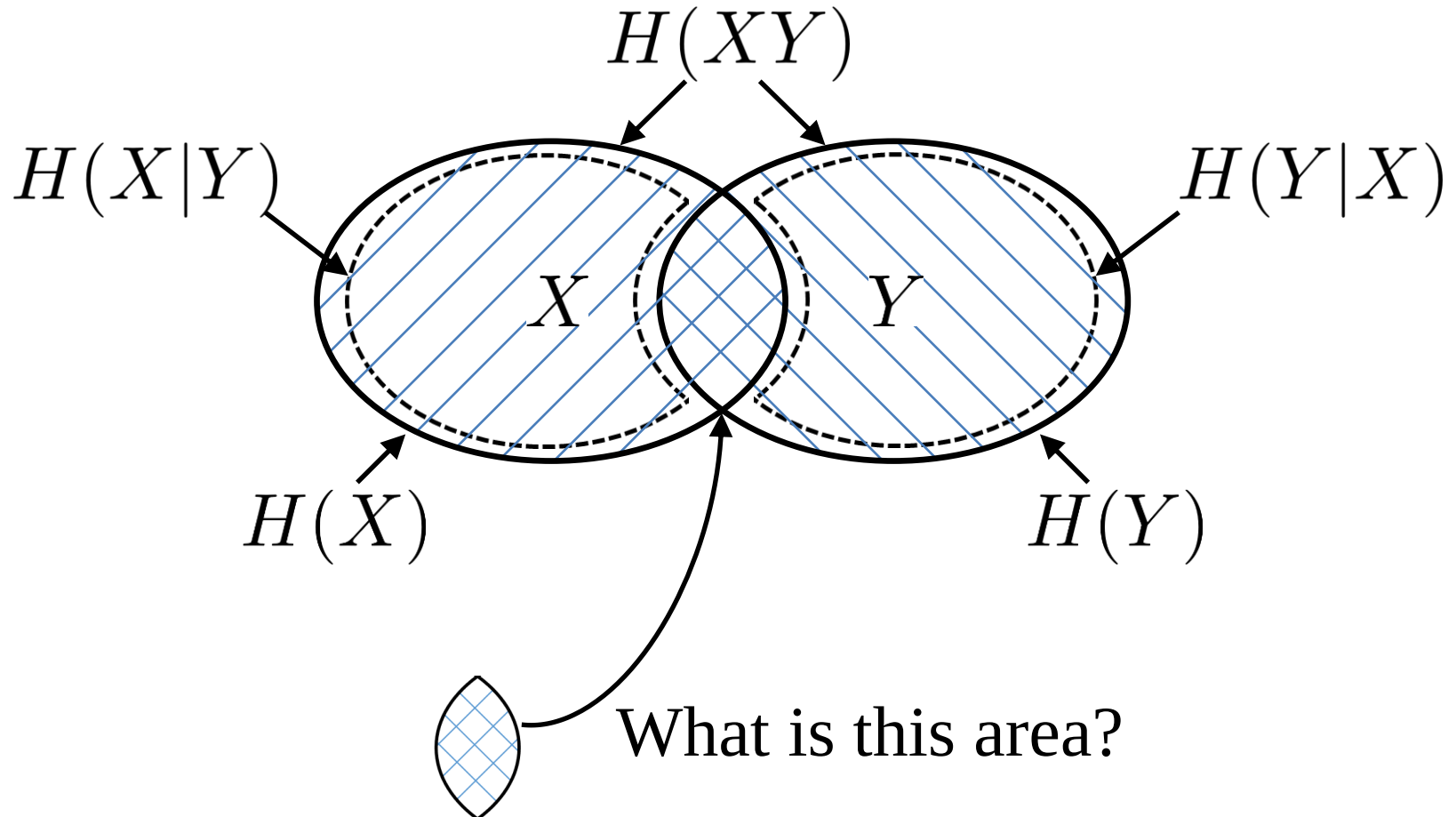
$$\iff H(XY) = H(X) \iff H(Y|X) = 0$$

- Q: $H(X|Y) \stackrel{?}{=} 0$
- $X \perp Z \implies H(X + Z|X) = H(Z|X) = H(Z)$



- From Entropy to Lossless Compression
- Joint Entropy and Conditional Entropy
- Mutual Information and Channel Capacity
- Differential Entropy, Mutual Information and Shannon's Formula

Mutual Information (1)



Mutual Information (2)

$$\begin{aligned} \text{Venn diagram} &= H(Y) - H(Y | X) \\ &= H(X) - H(X | Y) \\ &\stackrel{(a)}{=} H(X) + H(Y) - H(XY) \end{aligned}$$

- Reduction of Y 's uncertainty given X
- Reduction of X 's uncertainty given Y
- (a) using chain rule

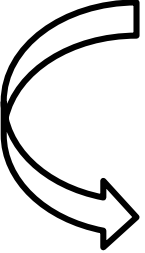
Mutual Information (3)

■ Define $I(X; Y) = H(X) + H(Y) - H(XY)$

1) $I(X; Y) = I(Y; X)$ (Mutual)

2) $I(X; Y) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{p_i p_j}$

3) $I(X; Y) = \sum_i p_i \sum_j p_{j|i} \log \frac{p_{j|i}}{\sum_i p_i p_{j|i}}$

 $X \longrightarrow \boxed{p_{j|i}} \longrightarrow Y$



Mutual Information (4)

- Properties:

- Non-negativity $I(X; Y) \geq 0$

- Three equivalent definitions

$$I(X; Y) = H(X) - H(X | Y)$$

$$I(Y; X) = H(Y) - H(Y | X)$$

$$I(X; Y) = H(X) + H(Y) - H(X, Y)$$

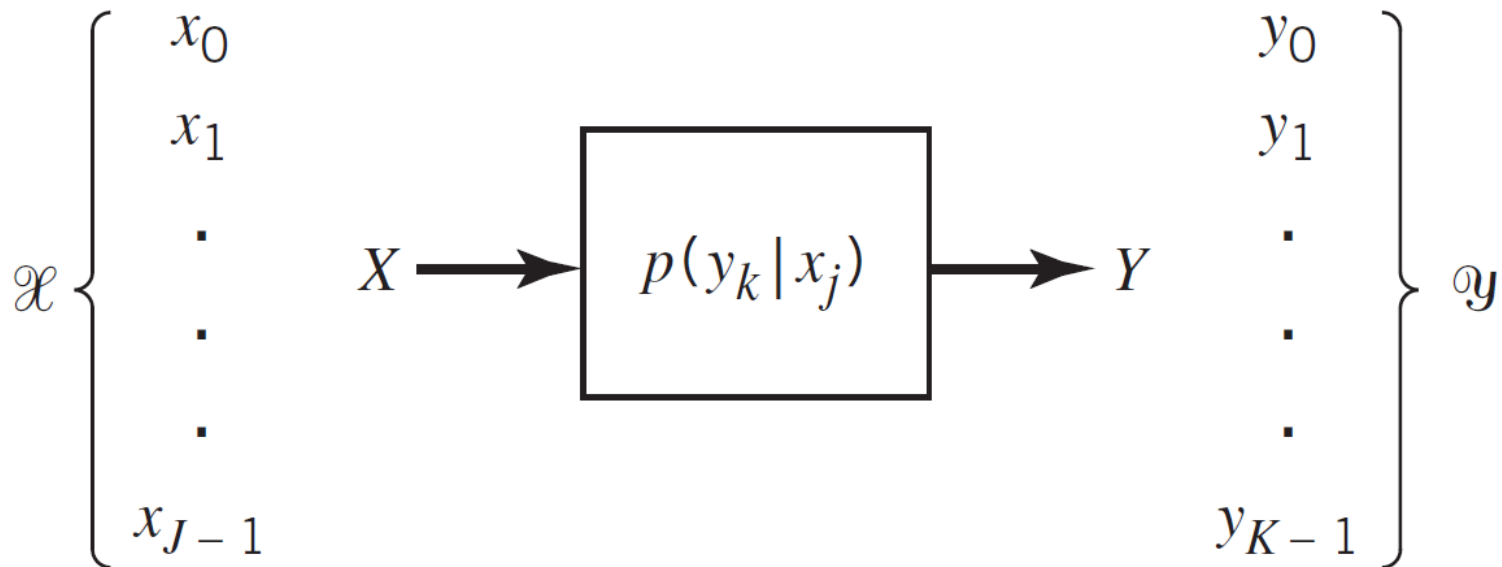
Discrete Memoryless Channel

■ Input alphabet

$$\mathcal{X} = \{x_0, x_1, \dots, x_{N-1}\}$$

■ Output alphabet

$$\mathcal{Y} = \{y_0, y_1, \dots, y_{M-1}\}$$



Channel Matrix

$$\mathbf{P} = \begin{bmatrix} p_{0|0} & p_{1|0} & \dots & p_{M-1|0} \\ p_{0|1} & p_{1|1} & \dots & p_{M-1|1} \\ \vdots & \vdots & & \vdots \\ p_{0|N-1} & p_{1|N-1} & \dots & p_{M-1|N-1} \end{bmatrix}$$

- Transition matrix with row-sum = 1



Channel Input and Output

- X : Input to the channel

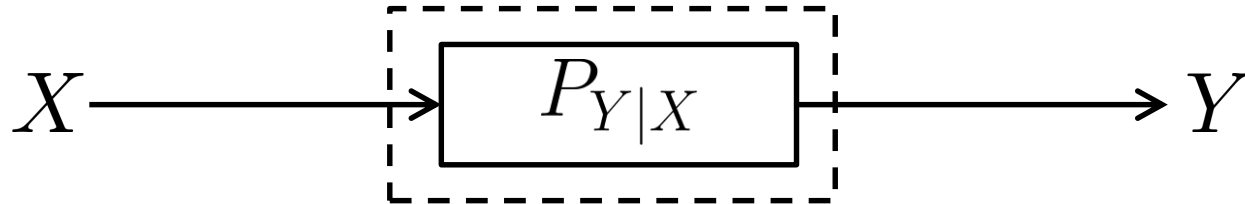
$$p_i = \Pr(X = x_i) \quad \text{for } i = 0, 1, \dots, N - 1$$

- Y : Output of the channel

- Joint Distribution

$$\begin{aligned} p_{i,j} &= \Pr(X = x_i, Y = y_j) \\ &= \Pr(Y = y_j \mid X = x_i) \Pr(X = x_i) \\ &= p_{j|i} p_i \end{aligned}$$

Channel Capacity (1)



Channel $P_{Y|X}$

- What we can control: $X[1], X[2], \dots, X[K]$
 - What we can observe: $Y[1], Y[2], \dots, Y[K]$
 - How many inputs $X[1 : K]$ are distinguishable?
 - ▶ Mutual information
- $$2^{KI(X;Y)}$$

Channel Capacity (2)

- The maximum number of distinguishable $X[1 : K]$ is

$$\mathcal{N} = 2^{K \max_{p(x)} I(X;Y)}$$

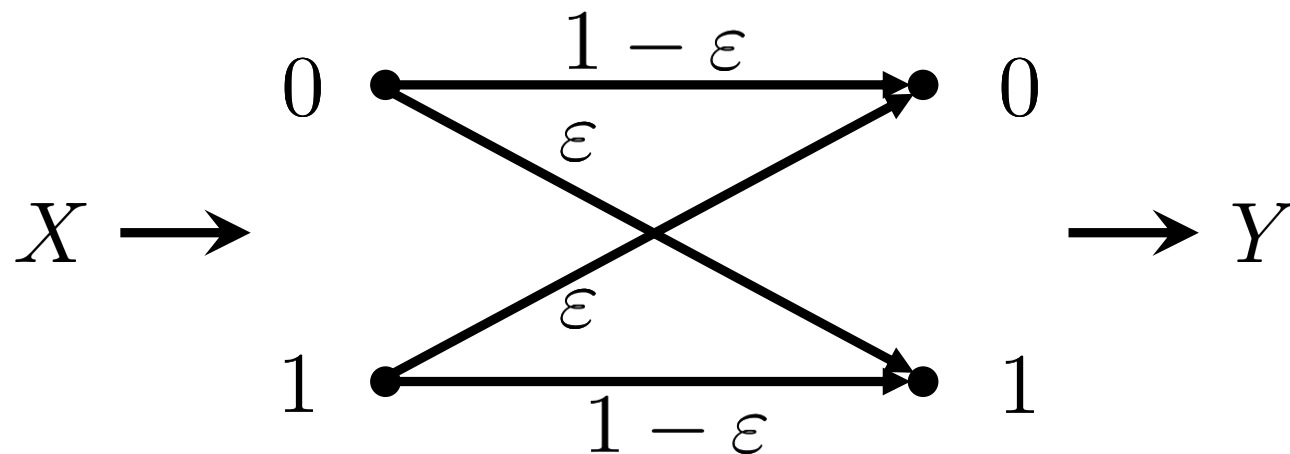
- Definition: **Channel Capacity** (per channel use) is the maximum number of distinguishable bits per X

$$C = \lim_{K \rightarrow \infty} \frac{1}{K} \log_2 \mathcal{N} = \max_{p(x)} I(X;Y)$$

Binary Symmetric Channel (1)

■ Binary Symmetric Channel (BSC)

$$X \sim \begin{pmatrix} 0 & 1 \\ 1-p & p \end{pmatrix} \text{ Bernoulli } (p)$$



■ Q: $I(X; Y) = ?$



Binary Symmetric Channel (2)

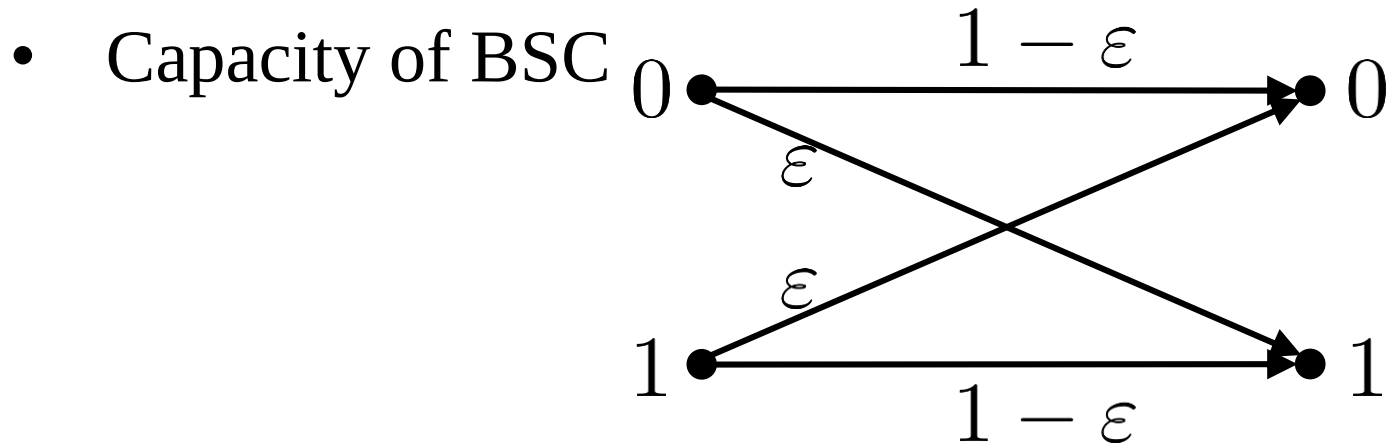
$$Y = X \oplus Z$$

$$Z \sim \text{Bernoulli}(\varepsilon)$$

$$Y \sim \text{Bernoulli}((1 - p)\varepsilon + p(1 - \varepsilon))$$

$$\begin{aligned} I(X; Y) &= H(Y) - H(Z) \\ &= -(\bar{p}\bar{\varepsilon} + p\varepsilon) \log_2(\bar{p}\bar{\varepsilon} + p\varepsilon) \\ &\quad - (\bar{p}\varepsilon + p\bar{\varepsilon}) \log_2(\bar{p}\varepsilon + p\bar{\varepsilon}) \\ &\quad + \varepsilon \log_2 \varepsilon + \bar{\varepsilon} \log_2 \bar{\varepsilon} \end{aligned}$$

Binary Symmetric Channel (3)



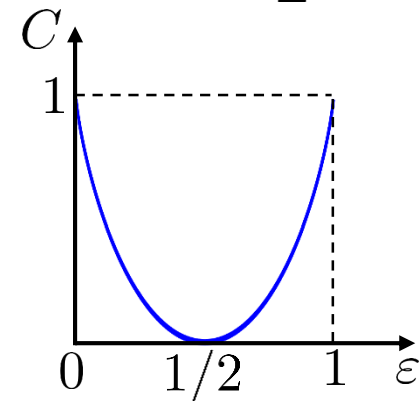
BSC(ε)

Constant

$$\max_p I(X; Y) = \max_p H(Y) + \varepsilon \log \varepsilon + \bar{\varepsilon} \log \bar{\varepsilon}$$

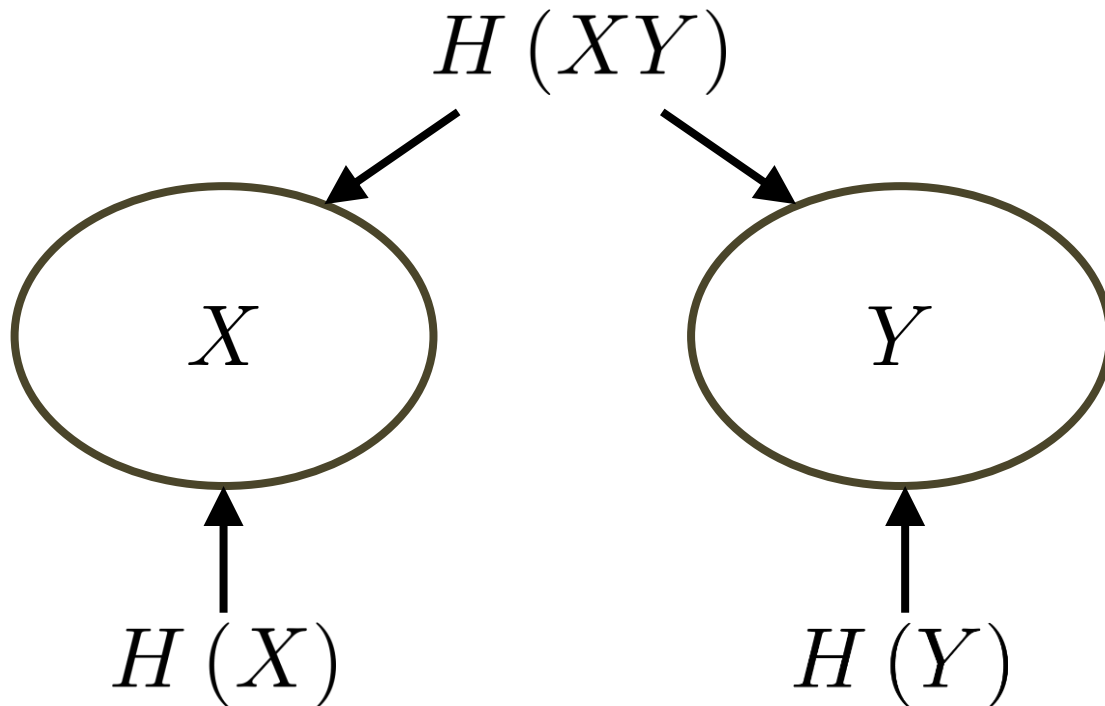
$$H(Y) \leq 1 \text{ and } H(Y) = 1 \text{ iif } \Pr\{Y = 0\} = \frac{1}{2}$$

$$\Rightarrow \begin{cases} p = \frac{1}{2} \\ C = 1 + \varepsilon \log \varepsilon + \bar{\varepsilon} \log \bar{\varepsilon} \end{cases}$$



Examples (1)

$$X \perp Y \iff p_{ij} = p_i p_j$$
$$\iff I(X; Y) = 0$$



- The uncertainty of X is not reduced by knowing Y



Examples (2)

$$Y = f(X) \Rightarrow I(X; Y) = H(f(X)) - H(f(X) | X) \\ = H(f(X))$$

■ If $\exists f^{-1}$, then $I(X; Y) = H(X) = H(Y)$

■ If $f(X) = X^2$, let

$$X \sim \begin{pmatrix} -2 & -1 & 1 & 2 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

then $I(X; Y) = H(f(X)) = H(X) - 1 = 1$



- From Entropy to Lossless Compression
- Joint Entropy and Conditional Entropy
- Mutual Information and Channel Capacity
- Differential Entropy, Mutual Information and Shannon's Formula



Extending to Continuous R.V.

■ Definition: For continuous r.v. X, Y

$$I(X; Y) = \iint_{yx} p(x, y) \log \frac{p(x, y)}{p_X(x)p_Y(y)} dx dy$$

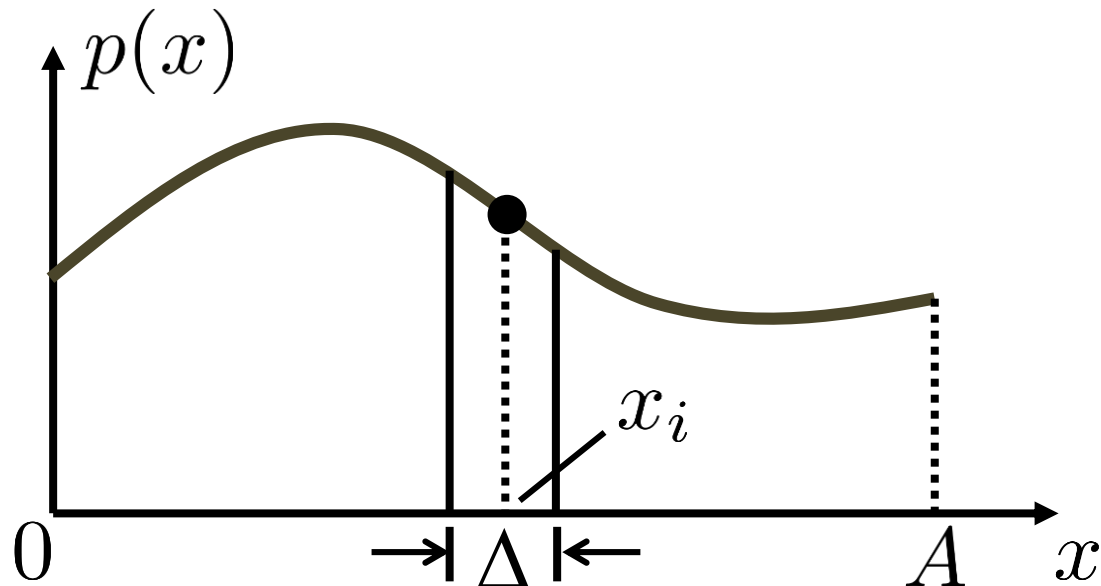
$$\begin{aligned} &= \boxed{- \int_x p_X(x) \log p_X(x) dx} \longrightarrow h(X) \\ &+ \boxed{\left(- \int_y p_Y(y) \log p_Y(y) dx \right)} \longrightarrow h(Y) \\ &- \boxed{\left(- \iint_{yx} p(x, y) \log p(x, y) dx dy \right)} \longrightarrow h(XY) \end{aligned}$$

Differential Entropy (1)

■ Definition:

$$h(X) = - \int_x p(x) \log p(x) dx$$

■ Q: Interpretation?



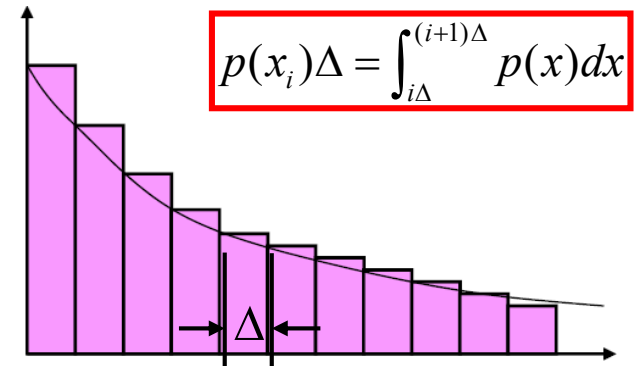
Differential Entropy (2)

■ Analogy

- ▶ Start from the entropy of discrete r.v.
- ▶ Substitute summation with integration

$$H(X) = - \sum_i p_i \log p_i$$

$$h(X) = - \int_{-\infty}^{\infty} p(x) \log p(x) dx$$



■ Rigorous derivation:

$$\begin{aligned} H(X) &= \mathbf{E}\{H(x < X \leq x + \Delta)\} = \mathbf{E}\left\{\log \frac{1}{p(x_i)\Delta}\right\} \\ &= - \int_{-\infty}^{\infty} p(x) \log p(x) dx + \lim_{\Delta \rightarrow 0} \log \frac{1}{\Delta} \end{aligned}$$

Differential Entropy (3)

- Decomposition of entropy:

$H(X) = -\int_{-\infty}^{\infty} p(x) \log p(x) dx$	$+ \lim_{\Delta \rightarrow 0} \log \frac{1}{\Delta}$
Differential Entropy: different among cont. random variables	Infinite term: constant for any cont. random variables

- We only care about the relative uncertainty of different cont. r.v.. Thus we define

$$h(X) = -\int_{-\infty}^{\infty} p(x) \log p(x) dx$$

Examples

■ $X \sim \mathcal{U}([-A/2, A/2])$

$\Rightarrow h(X) = \log A$

$X \sim \mathcal{N}(0, \sigma^2)$

■ $\Rightarrow h(X) = -\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} \right) dx$

$$= \log(e) \times \frac{1}{2\sigma^2} \times \sigma^2 + \log \left(\sqrt{2\pi\sigma^2} \right)$$
$$= \frac{1}{2} \log (2\pi e \sigma^2)$$



Properties of Differential Entropy

■ If $|X| \leq \frac{A}{2}$, then

$$h(X) \leq \log A$$

■ If $\mathbb{E}(X^2) \leq \sigma^2$, then

$$h(X) \leq \frac{1}{2} \log (2\pi e \sigma^2)$$

■ $h(aX) = h(X) + \log |a|$

Mapping of cont. R.V. can change differential entropy

Bijection of disc. R.V. preserves entropy



Joint and Conditional Differential Entropy

■
$$h(XY) = - \iint_y p(x, y) \log p(x, y) dx dy$$

■
$$h(Y | X) = - \iint_{yx} p(x, y) \log p_{Y|X}(y | x) dx dy$$

■ Chain's rule:
$$\underbrace{h(XY)} = \underbrace{h(X)} + \underbrace{h(Y|X)}$$

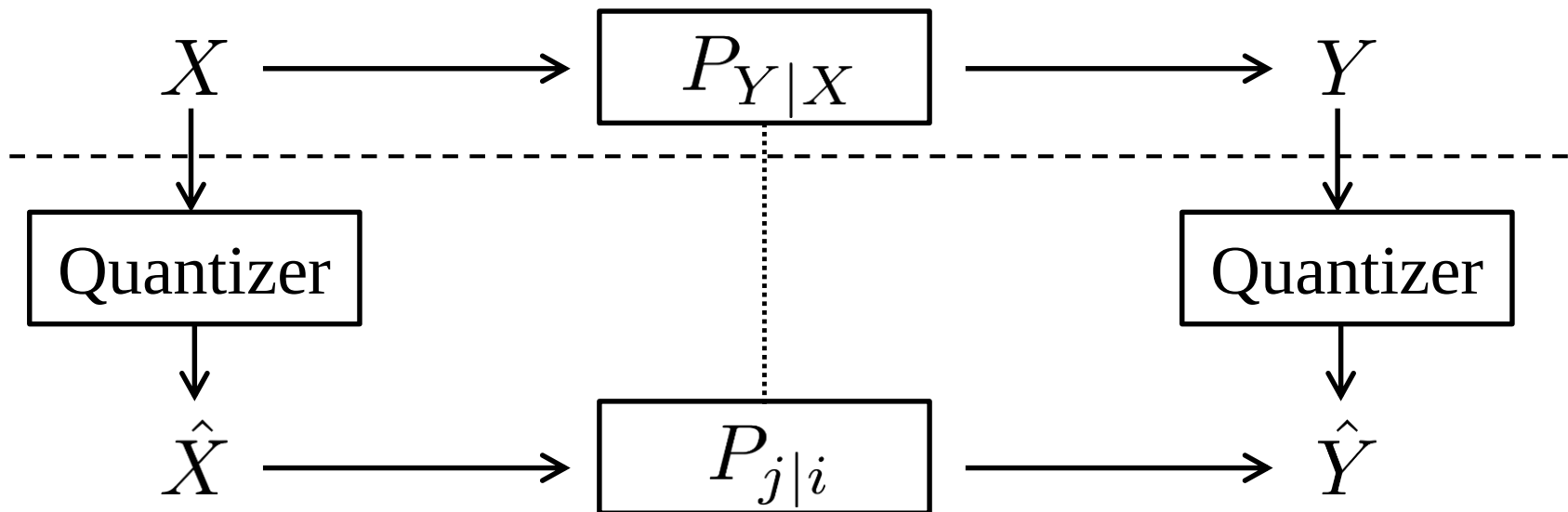
$$H(\hat{X}\hat{Y}) + 2(\log A - M)$$

$$H(\hat{Y}|\hat{X}) + \log A - M$$

$$H(\hat{X}) + \log A - M$$

Mutual Information of cont. random variables

$$\begin{aligned}
 I(X; Y) &= h(X) + h(Y) - h(XY) \\
 &= H(\hat{X}) + \log A - M + H(\hat{Y}) + \log A - M \\
 &\quad - [H(\hat{X}\hat{Y}) + 2(\log A - M)] \\
 &= H(\hat{X}) + H(\hat{Y}) - H(\hat{X}\hat{Y})
 \end{aligned}$$



An Example (1)

■ Gaussian channel

■ Power constraint: $\mathbb{E}\{X^2\} \leq P$

■ $\max_{p(x)} I(X; Y) = \max_{p(x)} h(Y) - \boxed{h(Z)}$
Constant, $\frac{1}{2} \log(2\pi e \sigma^2)$

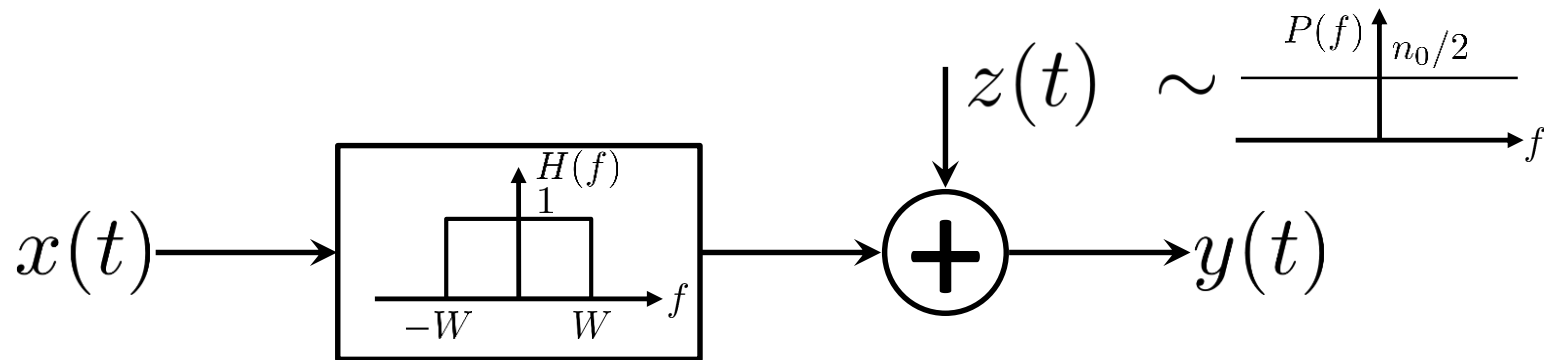
■ $\mathbb{E}(X^2) \leq P \Leftrightarrow \mathbb{E}(Y^2) \leq P + \sigma^2$
 $\Rightarrow h(Y) \leq \frac{1}{2} \log(2\pi e (P + \sigma^2))$ and " $=$ " holds iff

$$Y \sim \mathcal{N}(0, P + \sigma^2) \Leftrightarrow X \sim \mathcal{N}(0, P)$$

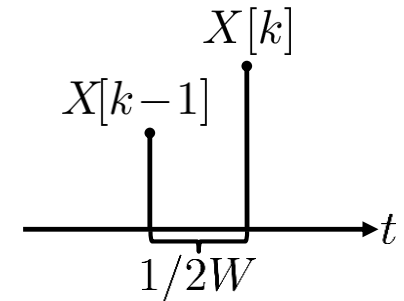
$$C = \frac{1}{2} \log \left(1 + \boxed{\frac{P}{\sigma^2}} \right) \rightarrow \gamma : \text{Signal-to-Noise Ratio (SNR)}$$

Band-limited AWGN Channel (1)

AWGN: Additive White Gaussian Noise



$$x(t) = \sum_{k=-\infty}^{\infty} X[k] \operatorname{sinc} \left(t - \frac{k}{2W} \right)$$



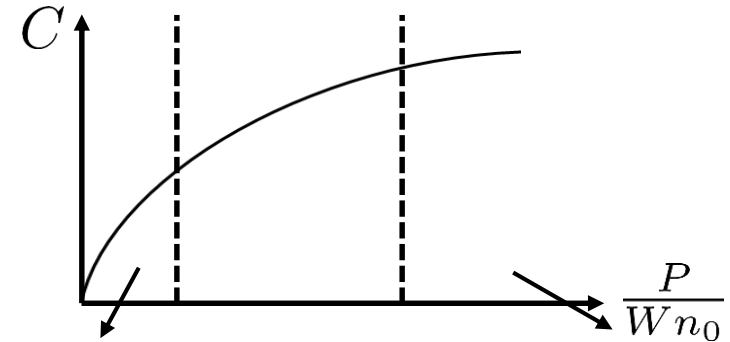
$$C = 2W \times \frac{1}{2} \log \left(1 + \frac{P}{W n_0} \right) \quad (\text{bps})$$

$$= W \log \left(1 + \frac{P}{W n_0} \right) \dots$$

Shannon's Formula

Band-limited AWGN Channel (2)

$$C = W \log \left(1 + \frac{P}{W n_0} \right)$$

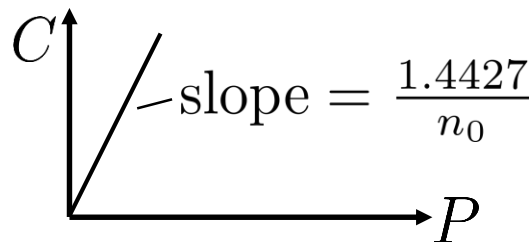


low SNR regime high SNR regime

Low SNR regime:

Taylor's Expansion $\rightarrow$

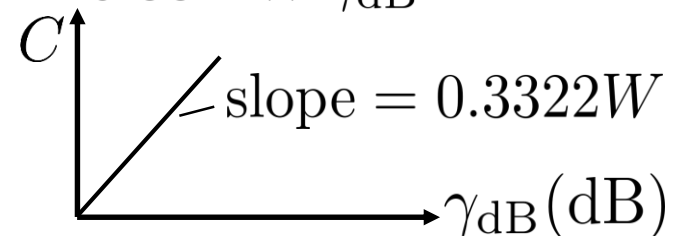
$$C \approx \cancel{W} \log_2 e \times \cancel{\frac{P}{W n_0}} = 1.4427 \times \frac{P}{n_0}$$



- 1) Power Matters
- 2) Bottleneck: Power

High SNR regime:

$$\begin{aligned} C &\approx W \log_2 \gamma \\ &= \frac{W}{10} \log_2 10 \gamma_{\text{dB}} (\text{in dB}) \\ &= 0.3322 W \gamma_{\text{dB}} \end{aligned}$$



- 1) Bandwidth Matters
- 2) Bottleneck: Bandwidth

Band-limited AWGN Channel (3)

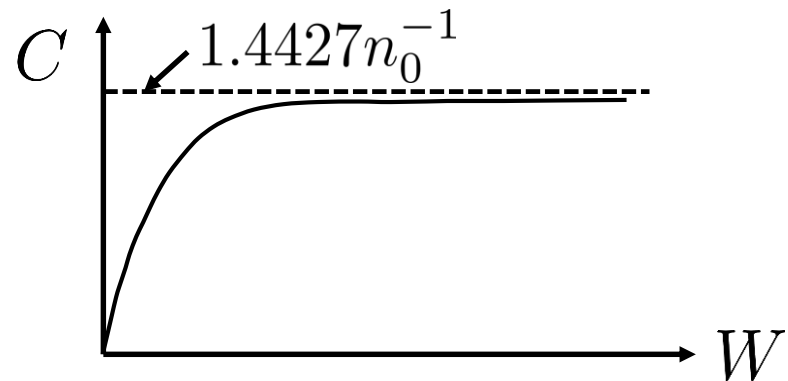
- Power Efficiency

$$\eta_P = \frac{C}{P} \quad (\text{Data Rate per Unit Power})$$

- Energy Efficiency

$$\eta_E = \frac{\# \text{ bits}}{E} = \frac{CT}{PT} = \frac{C}{P} = \eta_P \quad (\text{bit/Joule})$$

$$\eta \leq 1.4427n_0^{-1}$$



- Q: How to achieve $\eta_{\max} = 1.4427n_0^{-1}$?



Band-limited AWGN Channel (4)

- Policy 1: Ultra Wide Band (UWB)

$$W \rightarrow \infty, \eta \rightarrow \frac{1.4427 \times \cancel{P}}{\cancel{P} n_0} = 1.4427 n_0^{-1}$$

- Policy 2: Lazy Scheduling

$$T \rightarrow \infty, \eta = \frac{WT \log \left(1 + \frac{E}{WT n_0} \right)}{E}$$

$$\rightarrow \frac{\cancel{WT} \times 1.4427 \frac{\cancel{E}}{\cancel{WT} n_0}}{\cancel{E}} = 1.4427 n_0^{-1}$$

Green Communications



Thank you !

(Q&A)

Reference (Introductions):

[1] S. Haykin, *Digital Communication Systems*, Chap 1

[2] B. Forouzan, *Data Communications and Network*, Chap 1 & 2

Reference (Information Theory):

[1] S. Haykin, *Digital Communication Systems*, Section 5.1-5.10